# lsqvce 5.3 — User Manual

least-squares adjustment with variance component estimation

built from the sources; the single documentation authority

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# 1 Overview

**lsqvce** is a MATLAB toolbox for least-squares adjustment with variance component estimation, built for satellite gravity recovery (GRACE / GRACE-FO) and equally at home on a levelling network. Version 5 (September 2026) is the result of the R5 review track: one accumulator, one stochastic-model estimator, one loop with three policies, one user surface.

## 1.1 What it does

- **Adjusts** a linear or linearised Gauss–Markov model with a common parameter block and per-system, per-day local blocks eliminated on the fly (the block-arrow structure of satellite gravimetry), on one accumulator with a QR or normal-equation backend.
- **Estimates the stochastic model** of every observation stream jointly (Ellmer 2018): a spectral shape per pool group, a level per stream, optional per-arc variance factors, by Hutchinson-probed redundancies – `Estimate="noise"`.
- **Regularises by declaration**: priors on common and on local parameters are pseudo-observation groups in your columns, counted in the redundancy, with their sigma fixed or estimated by Helmert / Koch–Kusche.
- **Diagnoses**: Baarda / Pope data snooping, redundancy numbers, reliability, the whiteness of every stream, a five-level verdict (`R.report`).
- **Robust** M-estimation (IRLS) and damped Gauss–Newton on the same loop.
- **Watches**: `Options.OnPass` fires after every pass with the state and the live accumulator; `R.plot` afterwards.

## 1.2 The three sentences you need

```
lsqvce_setup                                    % once
R = lsqvce.fit(data, Estimate="noise", Pool="obsType");   % the one verb
R.report                                        % the verdict
```

`data` is a plain struct (chapter 3); everything about *how* to run is a `lsqvce.Options` (chapter 4); everything that comes back is a `lsqvce.Results` (chapter 5). The theory behind every number is `lsqvce_theory.pdf` (chapter 11).

## 1.3 Reading order

New here: chapter 2 (install), then `demo_workflow` next to chapter 3, then chapter 4 when you want to change a knob. Migrating from 4.x: chapter 12 first.

# 2 Installation

## 2.1 Requirements

MATLAB R2024a or later (developed and gated on R2026a). The Parallel Computing Toolbox is optional (`Parallel.Mode="parfor"|"spmd"`, `Parallel.DistributeDays`); without it the toolbox runs serially and says so once per run.

## 2.2 Install

```
cd <toolbox root>          % the folder holding +lsqvce
lsqvce_setup               % or lsqvce_setup('install')
```

This adds exactly two path entries (the root with the `lsqvce.*` packages, and `examples/`), builds the Help browser index so that `doc lsqvce` opens this manual, saves the path, and prints a check report. Not added by design: `dev/tests` (the suites manage their own path), `template/` (you copy it into your project), `dev/`.

## 2.3 Check

```
lsqvce_setup('check')      % version, path entries, doc index, whether lsqvce.fit resolves
```

## 2.4 Uninstall

```
lsqvce_setup('uninstall')  % removes the two path entries, saves the path, deletes the
                           % Help browser database it built; the folder itself is untouched
```

## 2.5 Layout of the toolbox folder

| folder / file | what |
|---|---|
| `+lsqvce/` | the package: user tier (`fit`, `Options`, `Results`, `report`, `layout`), `+engine`, `+kernel`, `+util` |
| `examples/` | the demos (`demo_workflow`, `demo_fit`, `demo_grace`, `demo_gravity`, `demo_gravfire`, ...) |
| `template/` | `gravfire_data.m` (a GRACE bundle to the data struct), `lsqvce_template.m` |
| `html/` | this manual (`lsqvce_manual.html`) and one page per public unit (`doc lsqvce.<name>`) |
| `lsqvce_theory.pdf`, `lsqvce_manual.pdf` | the theory and this manual as PDF |
| `RELEASE_NOTES.md` | what changed, with the migration table |
| `dev/` | the development record (`HANDOVER.md`), the test suites, the documentation sources |

# 3 Declaring the data

This chapter is the binding data contract of the toolbox (formerly the file `DATA_CONTRACT.md`, retired 2026-09-04: the Manual is the single authority). Everything the toolbox knows about your problem enters through the `data` struct described here.

**Status.** Binding specification, **version 5** (adopted 2026-09-03, Review R5 level M5: the v4 SHAPES are unchanged — every v4 declaration still fits — and three things were ADDED, listed in §0a below; option names are those of `lsqvce.Options` v5, the v4 → v5 map is in `RELEASE_NOTES.md`). `lsqvce.fit` is live for all three data kinds, `Estimate = "none" | "noise"`, `Robust`, the nonlinear

form and `OnPass`. §9 shows the equivalent direct engine calls. This file is the source of the `fit` help text and of the HTML guide: one authority, no second copy.

Created 2026-08-31 (CEST, UTC+02:00) with assistance from Claude (Anthropic), model Claude Fable 5. Rewritten for v3 and again for v4 (scoped tables) the same day (Claude Fable 5).

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### 3.0.1 0a. The v5 delta (R5 M5): three additions, no shape changes

1. **`data.priors(k) = struct('group' | 'L', 'x0', 'sigma', 'from', 'name', 'estimate')`** — declared pseudo-observation groups `L·x ~ x0` with weight `1/sigma²` on the COMMON parameters, folded into the solve AND into the diagnostic sweep (they carry a redundancy). A free network's datum is the prior `L = B'` on the null space: the exact minimum-constraint solution, for any weight, because the data's range and the null space are orthogonal (`lsqvce:fit:badPrior` on malformed fields). A Kaula rule or a background model is the same declaration with a scientific `L`. With `estimate = "scale"` (R5 M7 7b) the group has its OWN variance component: under `Estimate="noise"` its sigma is updated every pass by the Helmert / Koch–Kusche rule ($\Omega_k = \|(L\,x - x0)/\sigma\|^2$, $r_k = p - \mathrm{tr}(L\,N^{-1}\,L')/\sigma^2$, $\sigma^2 \leftarrow \sigma^2 \cdot \Omega_k / r_k$; the fixed point $\Omega_k = r_k$ is "the prior's weighted residual sum equals its redundancy"), so $\lambda = s0^2/\sigma_k^2$ is the regularisation the data support; `R.noise.priors` and `R.noise.priorHistory` record it. `from` is the pass at which the group becomes active (default 1; a single-pass run reports `lsqvce:fit:priorNotActive` and skips it). A group is declared EITHER by `group` — a regular expression on the NAME column of `commonParams` (`data.paramColumns.name`, default 2); fit builds L, one row per matched row per instance — OR by `L` ($p \times nc$ in the columns of A); never both (R5-D8 round 6). One `sigma` per group: the rows of L are the members, and several rows are what make `estimate="scale"` estimable (a single row has redundancy ≤ 1). 1b. **`data.systems(k).priors(l) = struct('group' | 'L', 'x0', 'sigma', 'from', 'name', 'estimate')`** (R5-D8 round 5/6) — the priors on system k's LOCAL parameters, the literal counterpart of `data.priors(j)`: one struct per GROUP, declared EITHER by **`group`** — a regular expression matched against the NAME column of the table (column 2; `data.paramColumns.name` to say otherwise); the toolbox builds L with one row per matched template row per instance — OR by **`L`** ($q \times nl$ in the columns of that system's B as **`lsqvce.layout(data)`** announces them, used as is); never both. Then `x0`, ONE `sigma` per group (relative weights inside a group go into L), `from`, `name` (default the pattern), `estimate = "none" | "scale"`. Every row must lie inside one day's local block (couple days through common parameters); the width of L is checked by name (`lsqvce:fit:badPrior`). Each group rides as its own observation stream `prior_<name>` of the system (pseudo-observations A = 0, B = L/σ, l = x0/σ, unit law), so accumulation, elimination, the sweeps, `residuals()` and the redundancy count treat it as data — and with `estimate = "scale"` its σ is estimated every pass of `Estimate="noise"` as that stream's level, the Helmert / Koch–Kusche factor the joint estimator computes for every stream: IDENTICAL behaviour to the common groups (`R.noise.model{k}.sigma` for stream `satKey|prior_<name>`). Common priors and every system's priors carry their own settings. THE RULE (R5-D8): the parameters and their columns are yours (tables, `buildDay`, plots, `toTable`); regularisation is the toolbox's (mechanism, redundancy, activation, VCE per group); the interface between them is an operator in your columns, announced by `lsqvce.layout`, checked by width. `Options.Reg / RegLambda / RegMatrix` are gone. 1c. **`data.paramColumns = struct('start', c1, 'end', c2)`** (R5-D8 round 3) — names the columns of BOTH tables that hold a row's validity (`[MJD seconds]` pairs or fractional MJD; `[]` = always). A `'day'` / `'arc'` row is then instantiated only on the calendar days its validity overlaps (half-open day: start < d+1 and end > d); `'common'` / `'system'` rows are always present. nc and nl are derived accordingly, and `buildDay`'s 6th argument tells you which template rows exist on the day: `info.arcRows`, `info.dayRows`, `info.sysRows`, `info.commonDayRows` (indices into the tables, in block order). Without `paramColumns` every row is valid always — the v4 behaviour. This is how a one-day manoeuvre parameter and a "literal" row become the same mechanism as everything else.

2. **Per-stream policy on the covariance spec** — `covariance.<type>.estimate = "none" | "scale" | "law"` (default `"law"` under `Estimate="noise"`) and `.arcFactors` (logical, default the Options policy). A law whose `sigma` is a VECTOR (per-row weights) is a fixed structure: it can carry a level (`"scale"`) but not a spectral shape (`"law"` is refused, `lsqvce:fit:lawOnFixedStructure`). `Options.Pool` decides which streams SHARE one estimated spectral shape; every stream keeps its own level. Of a stream's per-arc factors only the **ratio** is identifiable — the level absorbs their mean. (R6 finding F-01, fixed in WP-98: eq. 6.5.28 multiplies by the factor the weighting used, so the ratio now settles across passes — 9.4 / 8.6 / 8.6 / 8.6 against a planted 9, where before it grew by 8.4x every pass. Runs with `arcFactors = false` are unchanged.)
3. **The per-pass inspection hook** — `Options.OnPass = @(ctx)` receives, after every pass of every iterative run, the pass state, the live accumulator, `ctx.data` (this declaration as given) and `ctx.processingOptions{k}` (system k's `processingOptions` as declared); return `false` to stop. Nothing else about `buildDay` or the tables changed.

Semantics made explicit in v5: a full SPD weight matrix returned by a batch source is whitened exactly (`chol(P)`), and per-observation diagnostics of a batch problem are reported in the WHITENED frame (`R.diag.v = sqrt(p).*v`; only their sums are frame-invariant).

### 3.0.2 0b. The v5.3 delta (R6 WP-90): the error information travels with the day

What the preprocessing knows about the accuracy of every observation — a per-epoch standard deviation, per-axis standard deviations, or a full 3 × 3 covariance per epoch — is DATA, and it comes back from `buildDay` WITH the day, as **`D.Cep`** (§3.4, §7), for ANY stream: `nEpoch × 1` for a scalar stream, `nEpoch × d` or `nEpoch × d × d` for a multi-axis one. The axis count is derived from it. The covariance spec (§5) declares the TEMPORAL law only — `diag` (white), `arma`, `psd`, `acf` — and the toolbox composes C = D (R_t ⊗ I_d) D per arc. Retired with this: the `separable` type (it was `arma` + Cep), the `epochCovariance` hook (an unknown field now), and a static `Cep` on the spec (refused when the day carries one: one fact, one authority). Nothing else about the declaration changed.

### 3.0.3 0. The v4 principle: state nothing the data already states — and scope, don't number

The contract declares **structure**, never **counts**, and since v4 it declares **scope, never numbering**: column 1 of both parameter tables is a **scope token**, the rows are **templates**, and the toolbox expands them over the calendar. Every count is derived:

| quantity | derived from |
|---|---|
| nc (common unknowns) | the expansion of `data.commonParams` (`'common'` rows once + `'day'` rows per calendar day with arcs) |
| nl (local unknowns) | the expansion of `localParams` (`'arc'` rows per (day, arc) + `'day'` rows per day with arcs + `'system'` rows once); **not declared → nl = 0** |
| observation types | `fieldnames(covariance)` |
| u (batched nonlinear) | `numel(x0)` |

The v2 count fields are gone (v3), and so is v3's hand-numbering split (`localsPerArc` XOR free `localParams`): **one mechanism**, the scoped table. Every column offset reaches `buildDay` through its 6th argument (`info`, §6.3) — no offset is ever computed in user code. The driver was the GRACE parameterization: arc-wise ICs, day-wise ACC scale/bias/drift, per-arc KBR empiricals — three scopes, one table.

### 3.0.4 1. The mental model in one paragraph

You describe **one adjustment problem** with a plain MATLAB struct called `data`. It answers three questions: *what is estimated for everybody* (`data.commonParams`, one row per common unknown), *which days are processed* (`data.days`), and *which observing systems deliver data* (`data.systems`, one entry per satellite or instrument). Each system points to **one function you write**, `buildDay`, that returns the design matrices and observations of a single day. Everything else — reading days in the right order, cutting them into arcs, whitening with your covariance model, iterating the variance components — is the toolbox's job:

```
opts = lsqvce.Options(Estimate="noise");  % what to do
R    = lsqvce.fit(data, opts);           % the one verb
R.report                                 % the verdict
```

You never call `makeLoader` yourself; `fit` builds the loaders from `data`. `R.systems` (the existing Results field) records what ran: per system its id, nl, the loader closure, the day list and the name.

---

### 3.0.5 2. The `data` struct — field by field

`data` is a **scalar struct**. Field names are **case-sensitive** and use lowerCamelCase. Unknown field names are an **error** (`lsqvce:fit:unknownField`), which is deliberate: a typo in a struct field would otherwise be silently ignored. The retired v2 names (`common`, `numCommon`, `numLocal`, `observationTypes`, `systemId`, `cache`, `loader`, the plot hooks) are unknown fields now — the error names the field, so a v2 declaration fails loudly, not subtly.

#### 3.0.5.1 2.1 `data.commonParams` — the shared unknowns (required)

| field | type / size | meaning |
|---|---|---|
| `commonParams` | cell, R×C (C ≥ 1) | one row per common **template**. **Column 1 is the scope token**: `'common'` (one column of A, always) or `'day'` (one column per calendar day that carries at least one scheduled arc in at least one system). Columns 2…C are *your* convention (degree, order, epoch, …). nc is **derived** from the expansion; the table is piped to `buildDay` as its 3rd argument. `'arc'` is forbidden here — arc sets are system-specific (`lsqvce:fit:badScope`). |

**Binding column layout of A:** the `'common'` rows first (table order), then one day block per calendar day (calendar order — the sorted union of every declared day list). `buildDay` receives this day's block position as `info.commonDayOffset` / `info.commonDayCount` (§6.3). Minimal table when you have nothing to say: `[repmat({'common'}, nc, 1), num2cell((1:nc).')]`.

#### 3.0.5.2 2.2 `data.days` — the default schedule

| field | type / size | meaning |
|---|---|---|
| `days` | double, 1×D | day identifiers, e.g. `1:30` or `datenum` values cast to integers. Every system uses this list unless it carries its own `days`. Required unless every system has one. |

Days must be **numeric integers**: the loader cache is keyed on the number.

#### 3.0.5.3 2.3 `data.systems(k)` — one entry per observing system

`data.systems` is a **1×M struct array** (M ≥ 1). All entries have the same fields; fill an unused optional field with `[]`.

**Required — two fields**

| field | type / size | meaning |
|---|---|---|
| `buildDay` | function handle | your day builder, §3. |
| `covariance` | struct | one field per observation type holding a covariance spec (§5), e.g. `cov.kbr = struct('type','diag','sigma',0.2)`. **The fieldnames ARE the observation types** (their order is the processing order); `buildDay` is called once per (day, type). |

**Optional**

| field | type / size | default | meaning |
|---|---|---|---|
| `arcs` | double or fh | `1` | which arcs each day carries, §4. (`'fromDay'` is ENGINE-tier only.) |
| `localParams` | cell, R×C (C ≥ 1) | `{}` | the local **templates**, §6: column 1 is the scope token `'arc'` \| `'day'` \| `'system'`, columns 2…C your convention; nl is **derived** from the expansion. Passed to `buildDay` as its 4th argument. Not declared → **nl = 0** (§6.4). |
| `days` | double, 1×D | `data.days` | per-system schedule override. |

| field | type / size | default | meaning |
|---|---|---|---|
| `id` | double 1×1 **or** single char | the index k | system identifier, unique across systems (explicit ids and index defaults share one namespace). Longer labels go in `name`. |
| `name` | char / string | `''` | human-readable label used in reports. |
| `processingOptions` | struct | `struct()` | anything your `buildDay` needs (paths, force models); passed as its 5th argument. |

**Rule of thumb for beginners:** `buildDay` + `covariance` is a complete system. Add `arcs` when a day has more than one arc, `'arc'` template rows in `localParams` when arcs carry their own unknowns, the rest when you need it. (The v5.2 `epochCovariance` hook is gone — the per-epoch error information comes back from `buildDay` as `D.Cep`, §7.)

#### 3.0.5.4 2.4 A pitfall you will hit once: `struct()` with cell values

```
% WRONG: struct() expands cell values into a struct ARRAY
s = struct('localParams', {'bias','drift'});          % s is 1x2
% RIGHT: double braces, or assign field by field
s = struct('localParams', {{'bias';'drift'}});        % s is 1x1
s.localParams = {'bias'; 'drift'};                    % clearest
```

---

### 3.0.6 3. The `buildDay` contract

#### 3.0.6.1 3.1 Who calls it, and how often

`fit` calls `buildDay` **once per (day, observationType)** and cuts the returned day into arcs itself. Every pass over the data reads each day **twice** (once to accumulate the normal equations, once for the residuals), and the variance-component loop multiplies that by the number of passes. Two consequences:

1. **Determinism is mandatory.** Same arguments → byte-identical output. Random numbers, clocks, mutable globals are forbidden inside `buildDay`. (The residuals are formed as `v = l − A·x − B·y` from the second read; any difference lands in $\sigma_0$.)
2. **Cache expensive reads yourself.** The toolbox caches *one built day*; if the raw file read is the expensive part and serves several observation types, keep your own cache in the closure that binds `buildDay`.

#### 3.0.6.2 3.2 Signature — declare only what you need

`fit` passes as many of these six arguments as your handle **declares** (`nargin`):

```
D = buildDay(day, obsType)                                  % classic
D = buildDay(day, obsType, common, local, procOpt)          % parameter-driven
D = buildDay(day, obsType, common, local, procOpt, info)    % with the expansion info
```

| argument | what it is |
|---|---|
| `day` | one entry of the day list (numeric) |

| argument | what it is |
|---|---|
| `obsType` | one entry of `fieldnames(covariance)` (char) |
| `common` | `data.commonParams` |
| `local` | `localParams` of this system (or `{}`) |
| `procOpt` | `processingOptions` of this system |
| `info` | this day's expansion slice — every offset of A and B (§6.3); `[]` only when nothing is scoped |

A `varargin` builder receives all six.

#### 3.0.6.3 3.3 Return value — required fields

| field | size | meaning |
|---|---|---|
| `A` | m×nc | design matrix of the **common** unknowns, all observations of the day |
| `B` | m×nl | design matrix of the **local** unknowns; may be `sparse` (recommended when arc-wise) |
| `l` | m×1 | **observed minus computed** (the reduced observation). Not the raw observation. |
| `t` | m×1 | epoch of each row in **seconds** (the whitener needs the lags) |

Return **raw, unwhitened** matrices. Whitening happens per arc inside the toolbox from your covariance spec, and the VCE loop reweights there too — a builder that scales its own rows would be reweighted twice.

#### 3.0.6.4 3.4 Return value — optional fields

| field | size | meaning |
|---|---|---|
| `arcIdx` | cell, 1×nArc | row-index vector of each arc (lengths may differ). Absent → the whole day is one arc. |
| `arcId` | double, 1×nArc | the **identity** of each arc, when the schedule is ragged (§4.2). Must name every `arcIdx` entry exactly once with unique positive integers. |
| `Cep` | nEpoch×1, nEpoch×d or nEpoch×d×d | **the error information of this day** (§7): the per-epoch standard deviation of a scalar stream, the per-axis standard deviations, or the full per-epoch covariance of a multi-axis stream. nEpoch·d must equal the row count; the axis count d is derived from it. |

| field | size | meaning |
|---|---|---|
| `nAxis` | 1×1 | the axis count of a multi-axis stream WITHOUT `Cep` (rows epoch-major, m = nEpoch·nAxis, §7). Derived from `Cep` when that is present; given as well, it must agree. |

#### 3.0.6.5 3.5 What is checked, and the error you get

| error id | meaning |
|---|---|
| `lsqvce:makeLoader:buildDay` | your `buildDay` threw; the original error is attached as the cause |
| `lsqvce:makeLoader:badDay` | a required field (`A`, `B`, `l`, `t`) is missing |
| `lsqvce:makeLoader:dayShape` | `A`, `B`, `l`, `t` do not share a row count |
| `lsqvce:accum:arcBadNc` / `arcBadNl` | `A` or `B` has the wrong number of columns — the derived nc/nl are enforced against **every** day |
| `lsqvce:makeLoader:badArc` | a positional arc id is out of range (static schedule declared wrongly), or an arc of a multi-axis day does not hold whole epochs |
| `lsqvce:makeLoader:badArcId` | `arcId` does not name every `arcIdx` entry exactly once |
| `lsqvce:makeLoader:badDayCep` | `Cep` is not nEpoch×1 / nEpoch×d / nEpoch×d×d, or nEpoch·d is not the row count |
| `lsqvce:makeLoader:axisConflict` | `nAxis` disagrees with `Cep` (or with a `d` on the spec), or does not divide the row count |
| `lsqvce:makeLoader:cepTwice` | the covariance spec carries a `Cep` AND the day returns one |
| `lsqvce:covModel:cepOnVar` | a `var` law (its own cross-axis structure) with a `Cep` |
| `lsqvce:makeLoader:noCep` | *(warning, once per run)* a multi-axis day without `Cep`: the axes are weighted by the temporal law alone (D = I) |

These are raised **while reading**, by the engine, and keep their engine ids. Errors in the *declaration* itself are raised by `fit` before any data is read and carry `lsqvce:fit:` ids (`unknownField`, `noSystems`, `noCommon`, `missingField`, `localCount`, `noDays`, `badId`, `duplicateSystem`, `builderKind`, `badArcs`, and every `makeLoader` construction id relabelled, e.g. `lsqvce:fit:badScope`).

---

### 3.0.7 4. Arc schedules

#### 3.0.7.1 4.1 Static — every day looks the same

```
s.arcs = 4;            % arcs 1..4 on every day, every observation type
s.arcs = [1 2 4];      % these ids on every day
s.arcs.kbr = 2;  s.arcs.pos = 4;    % per observation type
```

Arcs are then addressed **by position** in `arcIdx`; an id beyond `numel(arcIdx)` is an error, which catches a mis-declared schedule.

#### 3.0.7.2 4.2 Ragged — days differ (gaps, manoeuvres, short months)

Declare a **resolver** that returns the arc ids of a day *without reading data*:

```
s.arcs = @(day, obsType) myArcTable{day};      % 1xA vector of ids for that day
```

and let `buildDay` return `arcId` naming the arcs it actually carries. Arcs are then addressed **by id**; an id that a day does not carry is treated as a **data gap** and skipped — no padding, no empty arcs.

(The v2 string form `arcs = 'fromDay'` is not part of this contract: it reads data before the run starts and cannot be combined with `localsPerArc`. It survives as an ENGINE-tier convenience on `lsqvce.engine.makeLoader`; `fit` rejects it with `lsqvce:fit:badArcs`.)

---

### 3.0.8 5. Covariance specs — the `covariance.(obsType)` structs

One struct per observation type; `type` selects the **temporal law** — how the errors of one stream are correlated down the epochs. The most common:

| `type` | fields | use for |
|---|---|---|
| `'diag'` | `sigma` (scalar std) | uncorrelated noise, e.g. `struct('type','diag','sigma',0.2)` |
| `'arma'` | `ar` `[1 a1 … ap]`, `ma` (default 1), `sigma` (innovation std) | coloured noise with a known filter |
| `'acf'` | `acf` `@(dt,opt)`, `sigma` | correlation given as a function of the lag |
| `'psd'` | `f` (nf×1, Hz), `S` (nf×1) | noise given as a one-sided PSD |
| `'var'` | `ar` (d×d×p), `sigma` (d×d) | vector-AR for multi-axis streams with cross-axis temporal correlation (no `Cep`) |

`sigma` is always a **standard deviation**, never a variance, and it is the LEVEL of the law. The accuracy of the individual observations — what varies from epoch to epoch — is not declared here: it comes back from `buildDay` with the day as `D.Cep` (§7), and the toolbox composes the two. The VCE loop scales and reshapes these specs at run time; you only give the starting point. Full reference: `doc lsqvce.engine.covModel`.

---

### 3.0.9 6. Local unknowns — the scoped `localParams` table

One mechanism: the table's rows are templates, column 1 says **how often** each row instantiates.

#### 3.0.9.1 6.1 The three scopes

```
s.localParams = [ ...
    [repmat({'arc'}, 6, 1),  num2cell((1:6).')];   % 6 ICs   -> once per (day, arc)
    {'arc','kbrBias'; 'arc','kbrCos1cpr'};         % 2 more  -> once per (day, arc)
    {'day', 'accScale'; 'day', 'accBias'};         % 2       -> once per day with arcs
    {'system', 'antennaOffset'}];                  % 1       -> once, full adjustment
```

| token | one block per … | typical use |
|---|---|---|
| `'arc'` | (day, arc) of this system | initial conditions, per-arc empiricals |
| `'day'` | day whose schedule carries ≥ 1 arc | ACC scale/bias/drift |
| `'system'` | whole adjustment | instrument constants; also the v3 free table (see migration) |

Requires `days` and a schedule resolvable **without reading data** (static or resolver) whenever `'arc'`/`'day'` rows are present; `'system'`-only tables need neither. nl is derived; declaring `numLocal` too is the conflict error.

**Binding column layout of B (per system):** per day (declared order) — the arc blocks in **sorted arc-id order** (each block = the `'arc'` rows, table order), then the `'day'` block — and after all days one trailing `'system'` block. A day whose schedule has no arcs gets **no blocks at all** (§6.4).

#### 3.0.9.2 6.2 Heterogeneous per-arc counts stay ENGINE-tier

A template has the same rows for every arc. If arcs genuinely carry *different* counts, that is the ENGINE-tier `lsqvce.engine.makeLoader(..., 'LocalsPerArc', @(day, arcId) n)` (the WP-52 plan) — deliberately not on the `fit` surface.

#### 3.0.9.3 6.3 Using `info` (buildDay's 6th argument)

```
info.arcId            % 1xA  ids of this day's arcs
info.offsetOf         % @(arcId) -> column offset of that arc's block in B
info.count            % 1xA  columns of each arc block (= number of 'arc' rows)
info.dayOffset        % 1x1  this day's 'day' block offset ([] if none)
info.dayCount         % 1x1  its width (0 if none)
info.sysOffset        % 1x1  the 'system' block offset
info.sysCount         % 1x1  its width
info.nl               % 1x1  total local count of the adjustment
info.nc               % 1x1  total common count
info.commonCount      % 1x1  width of A's leading 'common' block
info.commonDayOffset  % 1x1  this day's common day-block offset in A ([] if none)
info.commonDayCount   % 1x1  its width (0 if none)
```

Fill `B(rows, info.offsetOf(id) + (1:k))`, `A(:, info.commonDayOffset + (1:w))` — and **never compute an offset yourself**.

#### 3.0.9.4 6.4 Days without arcs, systems without locals

A calendar day on which **no system** schedules an arc gets no common day block (nc shrinks); a system day without arcs gets no arc and no `'day'` blocks (nl shrinks) — offsets stay gap-safe on ragged schedules and manoeuvre days. A system without local unknowns is still the **default**: declare no `localParams` and return `B = zeros(m, 0)` (the engine skips the pre-elimination; empty diagnostics per HANDOVER G-07).

---

### 3.0.10 7. The error information — `D.Cep` on the returned day

The covariance of a stream has two parts. The **temporal law** R_t (white, AR, a PSD) is declared in §5 and estimated by `Estimate="noise"`. The **per-epoch accuracy** D — what the preprocessing knows about each observation: a PPP covariance, a per-epoch sigma from a fit, a weight from an earlier screening — is DATA, and you return it with the day:

```
D.Cep = sigmaPerEpoch;                      % nEpoch x 1      scalar stream
D.Cep = [sx sy sz];                         % nEpoch x 3      per-axis sigmas (diagonal D)
D.Cep = permute(reshape(Q.', 3, 3, []), [3 1 2]);   % nEpoch x 3 x 3  full per-epoch covariance
```

The toolbox composes C = D (R_t ⊗ I_d) D per arc: every epoch is de-scaled by D^-1/2 (for d = 1 that is a division by the sigma), then the temporal whitener runs down the epochs of each axis. The axis count d is `size(Cep, 2)`; `nAxis` is derived from it. Rows of a multi-axis stream are **epoch-major** ($x_1\ y_1\ z_1\ x_2\ y_2\ z_2\ \ldots$), so nEpoch·d must equal the row count, and an arc must hold whole epochs. Under `Estimate="noise"` the estimator de-scales the residuals of every arc by the same D before it forms the lag statistics, so the estimated law is the law of the *normalised* errors — a scalar stream with a per-epoch sigma is handled exactly like a 3-axis one.

Rules:

- One authority: `Cep` lives on the day. A `Cep` on the covariance spec together with one on the day is refused (`cepTwice`); a `d` on the spec that disagrees with the day is refused (`axisConflict`).
- A multi-axis day **without** `Cep` (set `nAxis`) is composed with D = I and reported once per run (`lsqvce:makeLoader:noCep`) — the omission is distinguishable from a choice.
- `sigma` on the spec stays the LEVEL of the law; `Cep` is relative to it. Declare `sigma` 1 when `Cep` already carries the absolute accuracy.
- A `var` law carries its own cross-axis structure and takes no `Cep`.

---

### 3.0.11 8. Worked example — the GRACE parameterization: three scopes, one table

Arc-wise initial conditions, **day-wise** accelerometer scale/bias/drift, per-arc KBR empiricals — the case v3 could only hand-number, and the driver of v4:

```
% ---- the one function you write -------------------------------------
function D = buildDayGrace(day, obsType, common, ~, procOpt, info)
    raw = readGraceDay(procOpt.dataDir, day, obsType);   % your reader, deterministic
    D.A = shPartials(raw, common);                       % m x info.nc; the 'common' block
    D.l = raw.observed - raw.computed;                   % O - C
    D.t = raw.tSeconds;
    D.arcIdx = raw.arcRows;                              % cell, one row vector per arc
    D.arcId  = raw.arcIds;                               % this day's arc identities
    D.B = sparse(numel(D.l), info.nl);
    for a = 1:numel(D.arcId)
        off = info.offsetOf(D.arcId(a));
        D.B(D.arcIdx{a}, off + (1:info.count(a))) = icAndKbrPartials(raw, a);
    end
    if info.dayCount > 0                                 % this day's ACC block
        D.B(:, info.dayOffset + (1:info.dayCount)) = accPartials(raw);
    end
end

% ---- the declaration ------------------------------------------------
data.commonParams = [repmat({'common'}, 3717, 1), shParameterTable(60)];
data.days = 1:31;
cov.kbr = struct('type','arma','ar',[1 -1.6 0.9 -0.25],'ma',1,'sigma',1e-7);
cov.pos = struct('type','diag','sigma',0.02);             % fieldnames = the two types

s = struct();
s.buildDay = @buildDayGrace;
s.covariance = cov;
s.id = 'C';
s.name = 'GRACE-FO C';
```

```
s.arcs = @(day, obsType) arcSchedule('C', day);
s.localParams = [ ...
    [repmat({'arc'}, 6, 1), {'x0';'y0';'z0';'vx0';'vy0';'vz0'}];  % ICs per arc
    {'arc', 'kbrBias'; 'arc', 'kbrCos1cpr'};                      % KBR per arc
    [repmat({'day'}, 3, 1), {'accScale';'accBias';'accDrift'}]];  % ACC per day
s.processingOptions = struct('dataDir', '/data/gracefo');
data.systems(1) = s;

s.id = 'D';  s.name = 'GRACE-FO D';  s.arcs = @(day, obsType) arcSchedule('D', day);
data.systems(2) = s;

% nl per satellite = SUM over its days-with-arcs of (arcs*8 + 3) -- derived,
% gap-safe on manoeuvre days, never stated.

% ---- the run ------------------------------------------------------------
opts = lsqvce.Options(Estimate="noise", Pool="stream", ArcFactors=false, Parallel=struct('Mode',
R = lsqvce.fit(data, opts);
R.report
```

---

### 3.0.12 9. What `fit` does internally — the equivalent direct engine calls

```
L = cell(1, numel(data.systems));
for k = 1:numel(data.systems)
    s = data.systems(k);
    L{k} = lsqvce.engine.makeLoader(s.buildDay, ...
        ObservationTypes=fieldnames(s.covariance).', Covariance=s.covariance, ...
        SystemId=s.id, Name=s.name, Arcs=s.arcs, ...
        LocalParams=s.localParams, Scoped=true, CommonPlan=plan, ...  % fit builds `plan`
        Days=data.days, CommonParams=data.commonParams, ...           % over the global calendar
        ProcessingOptions=s.processingOptions);
end
R = lsqvce.engine.adjustBlock(L, data.days);                          % Estimate "none"
R = lsqvce.engine.adjustRobust(L, data.days, 'Influence', "huber");  % Robust "huber"
% Estimate "noise" is lsqvce.engine.adjustVCEjoint over an accumulator factory that
% lsqvce.fit builds (jointModelOpt + the block accumulator); use fit for it.
```

These are ENGINE-tier calls (experts); a user needs none of them. The ENGINE tier keeps the v2/v3 conveniences (`numCommon`/`numLocal` counts, free un-scoped tables with `Scoped=false` — the default — `LocalsPerArc` incl. the heterogeneous `@(day,arcId)` form, `'fromDay'`, `cache`, the plot hooks, raw 4-arg loader closures) — they left the USER surface, not the toolbox.

Field names map 1:1 to `makeLoader`'s Name-Value names (lowerCamelCase field ↔ UpperCamelCase name); `fit` performs exactly this mapping internally, plus the three derivations of §0.

---

### 3.0.13 10. Checklist before the first run

- ☐ `buildDay` returns `A`, `B`, `l`, `t` with one row count; `l` is O − C; nothing is whitened.
- ☐ Calling `buildDay` twice with the same arguments gives identical output.
- ☐ `t` is in seconds; the sampling matches the covariance spec you declared.
- ☐ `data.commonParams` rows carry the scope token in column 1; `size(A, 2)` must equal `info.nc` (use `zeros(m, info.nc)` and place blocks by the `info` offsets).
- ☐ Every observation type is a field of `covariance` — the fieldnames are the type list.
- ☐ Ragged schedule → resolver + `arcId`; static schedule → count/vector, no `arcId`.

- ☐ Local unknowns: scoped `localParams` templates, `B` filled from `info` offsets only; not declared → `B = zeros(m, 0)`.
- ☐ The error information: `Cep` on the day (nEpoch×1 for a scalar stream; nEpoch×d or nEpoch×d×d, rows epoch-major, for a multi-axis one); the spec declares the temporal law and its level only.
- ☐ No unknown field names in `data` (a typo is an error, on purpose — and so are the v2 count fields and, since v4, `localsPerArc`).

---

#### 3.0.14 11. The two in-memory kinds (no local unknowns, diagonal weights)

Not every problem streams. When A fits in memory and there are no arc-wise unknowns, `data` has no `systems` at all — `fit` tells the kinds apart by the fields present:

**Kind B — matrices**

| field | size | meaning |
|---|---|---|
| `A` | m×u | design matrix |
| `l` | m×1 | observed minus computed |
| `p` | m×1 | weights (diagonal of P), optional; absent = unit weights |

**Kind C — batched callback** (A too large for one matrix, or naturally batched)

| field | size | meaning |
|---|---|---|
| `batchFcn` | fh | `[A, l, p] = batchFcn(k)`, k = 1…nBatch — the `lsqvce.engine.adjust` contract |
| `numBatches` | 1×1 | nBatch |
| `numUnknowns` | 1×1 | u — **linear form only** |
| `x0` | u×1 | optional. With it the problem is **nonlinear**: `[A, w, p] = batchFcn(k, x)` returns the Jacobian at `x` and `w = l − f(x)`; solved by Gauss-Newton (`lsqvce.engine.solveNonlinear`). **u = numel(x0)** — `numUnknowns` must then be absent (`lsqvce:fit:uCount`) |

Rules: `opts.Estimate` must be `"none"` (variance components need the streams of a streaming system — `lsqvce:fit:modelKind`); `opts.Robust = "huber" | "igg3" | "bisquare"` switches to M-estimation (`RobustOpts` in Options); shapes are checked before anything runs (`lsqvce:fit:badMatrix`, `lsqvce:fit:missingField`, `lsqvce:fit:unknownField`). Both kinds run on the ONE accumulator as a local-free streaming system (batch k = day k), so `R.residuals()`, `R.contribution()`, `R.plot(..)` and `OnPass` work on them too; the per-observation diagnostics are in the whitened frame (`R.diag.v = sqrt(p).*v`).

What comes back is the same `Results`: `R.x`, `R.sigma0`, `R.coefficients()`, `R.report()`, `R.diag`, and for IRLS / Gauss–Newton the pass record on `R.history` (the final IRLS factors: `R.history(end).factor`) with `R.stopReason`.

```
data = struct('A', A, 'l', l, 'p', p);
R = lsqvce.fit(data);                                   % adjustBlock on the batch adapter
R = lsqvce.fit(data, Robust="huber");                   % adjustRobust (IRLS)
data = struct('batchFcn', @(k, x) myJacobian(k, x), 'numBatches', 20, 'x0', x0);
R = lsqvce.fit(data, MaxPasses=15);                     % adjustNonlinear (Gauss-Newton)
```

Stream names carry no meaning: the axis count of a stream comes from the day it returns (`Cep` or `nAxis`, §7), never from its name.

# 4 Options — the complete reference

`lsqvce.Options` is a typed value class: every knob has a type, a range and a default, checked at assignment (`MATLAB:validators:*` identifiers) or at construction (`lsqvce:Options:*`). It says *how to run*; everything about the data, including regularisation, is declared on `data` (chapter 3). Three equivalent spellings:

```
opts = lsqvce.Options(Estimate="noise", MaxPasses=6);   R = lsqvce.fit(data, opts);
opts.Noise.Grid = 2160;                                  % change one later (validated)
R = lsqvce.fit(data, Estimate="noise", MaxPasses=6);     % straight on fit
```

`disp(opts)` lists every knob, a * marking the defaults. The option space is **14 top-level knobs and 5 groups**; the count is pinned by a test so it cannot grow unnoticed.

## 4.1 What to estimate

**`Estimate`** — `"none"` (default) | `"noise"`. `"none"` is one generalised least-squares solve under the declared covariance laws. `"noise"` runs the joint variance-component estimator (Ellmer 2018, theory ch. 6): for every stream whose spec allows it (`covariance.<type>.estimate ~= "none"`), the spectral *shape* of its noise is estimated per pool group from Hutchinson-probed redundancies, one *level* per stream, and (with `ArcFactors`) one factor per arc; prior groups with `estimate="scale"` get their sigma by Helmert / Koch–Kusche in the same passes. The v4 words map as: `"ellmer"` → `Estimate="noise"`; `"vce"` → `Estimate="noise", Pool="stream", ArcFactors=false`; `"pooled"` → `Estimate="noise", Pool="all", ArcFactors=false`.

**`Pool`** — `"all"` (default) | `"obsType"` | `"stream"`. Which streams *share* one estimated spectral shape: all streams of the run; the streams of one observation type across systems (both satellites' KBR, say); or every (system, type) stream on its own. Every stream keeps its own level regardless. Fewer groups = more arcs per shape = a steadier estimate; more groups = more physics.

**`ArcFactors`** — logical, default `true`. Per-arc variance factors (theory 6.5.4): every arc of a stream gets a keyed factor on top of the stream's level, normalised to unit mean over the run. `false` = one level per stream (the Helmert factor of its arcs). Turn off when arcs are few or short.

**`Robust`** — `"none"` (default) | `"huber"` | `"igg3"` | `"bisquare"`. IRLS M-estimation on the same loop, every data kind: the influence function named here reweights the standardised whitened residuals with a MAD scale each pass; the final factors are `R.history(end).factor`. Not combinable with `Estimate="noise"` in one run (refused by name: run the robust pass with `Estimate="none"`).

## 4.2 The loop

**`MaxPasses`** — integer ≥ 1, default 10. The one pass budget of every iterative run (variance components, IRLS, Gauss–Newton). Exhausting it returns the last pass with `R.stopReason = "maxPasses"` and one warning.

**`MinPasses`** — integer ≥ 1, default 1. Passes run even if the tolerance is already met (GRAVFIRE's `minimumIterations`).

**`Tolerance`** — positive, default 1e-2. The stop rule of the variance-component loop: max $|\alpha^2 - 1|$ over frequencies and arcs below it; for Gauss–Newton the relative step in units of the formal error; IRLS uses `RobustOpts.Tol`.

**`Probes`** — integer ≥ 0, default 100. Hutchinson probes for the per-lag redundancies of the joint estimator; `0` = the exact route (measured to be the better default up to u ≈ 4000; the probe route is faster only for the dense inverse at small nc).

**`Seed`** — `[]` (default) or a double. RNG seed for the probes; recorded in `R.meta.seeds` so a run is reproducible.

## 4.3 Run control

**`Backend`** — `"qr"` (default) | `"normal"`. Square-root (Householder QR, rank refusal by `rank()`'s tolerance) or normal equations (Cholesky). Same numbers to rounding; QR is the safer default on ill-conditioned common blocks.

**`CacheDays`** — positive, default 1. Built days kept resident per system. A two-stage pass builds every day twice (accumulate + sweep) and a VCE run 2n+1 times; `Inf` keeps the calendar and builds each day once per run — your memory decision (a built type-day is m × nc doubles, ≈190 MB at degree 60 for a 6-h three-axis piece).

**`Verbose`** — logical or 0..2, default `false`. Screen messages: 0 silent, 1 progress, 2 detail. (Level 3 was merged into 2 in 2026-07; `lsqvce.util.verbosity` refuses it, and since v5.3 so does `Options` itself, by name.)

**`LogFile`** — string, default `""`. Mirror the messages to this file.

**`OnPass`** — `[]` (default) or `@(ctx)`. Your inspection hook, called at the end of **every pass** of every iterative run with one struct: `driver`, `pass`, `x`, `Cxx`, `sigma0`, `s0sq`; the loop state (noise: `C`, `Cgroups`, `sigma2`, `arcKeys`, `streams`, `sigma`, `white`, `whitePval`, `maxDev`, `skips`; IRLS: `factor`, `keys`, `scale`, `dx`; Gauss–Newton: `vPv`, `dxRel`, `lambda`); `acc` — the **live accumulator** (call `lsqvce.engine.residuals(ctx.acc, …)` or any engine function inside the hook); `data` — your declaration as passed to `fit`; `processingOptions{k}` — system k's, as declared. Plot, `drawnow`, save wherever you like. Return nothing to continue, `false` to stop after the pass (`R.stopReason = "userStop"`, the last pass returned with its covariance). A throwing hook is not swallowed: it arrives as `lsqvce:callUser:threw` with your error as the cause. This replaces the v4 `plotPostSolve` / `plotPostModel` hooks and `ShowSeconds` / `ExportDir`.

## 4.4 The groups

A group is a struct; assign a whole (partial) struct or one field. Partial structs are merged onto the defaults; an unknown field is refused (`lsqvce:Options:badGroupField`).

### 4.4.1 `Noise` — the estimated stochastic model

| field | default | meaning |
|---|---|---|
| `Grid` | 60 | frequencies (lags) of the estimated shape; must be ≥ the longest arc for an exact law (`"auto"` = the longest arc) |
| `Whitener` | `"toeplitz"` | how a PSD law whitens an arc: exact Toeplitz Cholesky, or an AR / ARMA filter fitted to it (`"ar"`, `"arma"`: O(m·p) instead of O(m²)) |

| field | default | meaning |
|---|---|---|
| `MeanType` | `"robust"` | the normalising mean of the arc factors: trimmed (`"robust"`) or plain |
| `Smooth` | 0 | moving-average half-width on the estimated PSD (0 = none) |
| `FreezeRedundancy` | 1 | a frequency whose redundancy is below this holds its value instead of being updated |
| `SkipThreshold` | 0.2 | the skip *rate* (arcs the sweep could not process / arcs of the stream) above which the pass is refused rather than silently thinned |

#### 4.4.2 `RobustOpts` — IRLS

| field | default | meaning |
|---|---|---|
| `Tuning` | `[]` | the influence function's tuning constant (its textbook default when empty: Huber 1.345, IGG3 (1.5, 2.5), bisquare 4.685) |
| `MaxIter` | 30 | IRLS passes |
| `Tol` | 1e-8 | relative change of x that ends the iteration |

#### 4.4.3 `Diagnostics` — the per-observation tests (`R.diag`, `R.report`)

| field | default | meaning |
|---|---|---|
| `Alpha` | 1e-3 | significance level of the outlier test |
| `Bonferroni` | `true` | correct α for the number of tests |
| `Beta` | 0.2 | power (1 − β) for the minimal detectable bias |
| `RMin` | 0.01 | redundancy number below which an observation is "untestable" |
| `RWeak` | 0.1 | below which it is "weakly controlled" |
| `Test` | `"auto"` | `"pope"` ($\sigma_0$ estimated) / `"baarda"` ($\sigma_0$ known) / `"mad"` (robust scale) / auto by the a-priori backstop |
| `RobustCheck` | `true` | run the MAD-scale backstop that flags a contaminated fit |
| `Store` | `"all"` | keep per-observation arrays for all / `"flagged"` / `"none"` |

| field | default | meaning |
|---|---|---|
| `NWorst` | 1000 | how many worst observations `R.report` keeps |

#### 4.4.4 `Parallel`

| field | default | meaning |
|---|---|---|
| `Mode` | `"serial"` | `"parfor"` or `"spmd"` with the Parallel Computing Toolbox; serial fallback with one warning without it |
| `DistributeDays` | `false` | the day loop of the accumulation over workers (bitwise-identical fold order) |
| `Threshold` | 8 | days below which distribution is not attempted |

#### 4.4.5 `GaussNewton` — problems with `data.x0`

| field | default | meaning |
|---|---|---|
| `TolF` | 1e-10 | relative change of the objective that ends the iteration |
| `Lambda0` | 1e-6 | initial Levenberg–Marquardt damping; a rejected step multiplies it |

### 4.5 Retired knobs

Regularisation is no longer an option (`Reg`, `RegLambda`, `RegMatrix`, `DatumB`, `Sigma0`, `Sparse` are gone): declare `data.priors` / `data.systems(k).priors` (chapter 6). The full v4 → v5 map is chapter 12.

## 5 The user tier: fit, Options, Results, report, layout

The help blocks of the five user-tier units, verbatim (`help lsqvce.<name>` shows the same text; `doc lsqvce.<name>` the same page). This chapter is generated from the code at build time.

### 5.1 lsqvce.fit

LSQVCE.FIT Fit a least-squares adjustment (optionally with VCE) to declared data.

```
LSQVCE.FIT  Fit a least-squares adjustment (optionally with VCE) to declared data.

  R = lsqvce.fit(data)                       every default (one GLS solve)
  R = lsqvce.fit(data, opts)                 a lsqvce.Options object
  R = lsqvce.fit(data, Name, Value, ...)     the Options properties, straight on fit
  R = lsqvce.fit(data, opts, Name, Value)    both: the pairs override opts

  THE ONE VERB. You describe the problem with a plain struct `data`
```

```
(manual, chapter 3), say what to do with lsqvce.Options (chapter 4), and
get a lsqvce.Results back (chapter 5):

    R = lsqvce.fit(data, Estimate="noise", Pool="obsType", OnPass=@myInspect);
    R.report;  R.plot("psd");  [C, L] = R.toTable(data);

THREE DATA KINDS, told apart by the fields of `data`, all on ONE accumulator:
  (A) streaming   data.commonParams, data.days, data.systems(k) with
                  buildDay / covariance / arcs / localParams / priors /
                  processingOptions; optional data.priors, data.paramColumns
  (B) in memory   data.A (m x u), data.l (m x 1), optional data.p (m x 1 weights
                  or m x m SPD weight matrix); optional data.priors
  (C) batched     data.batchFcn @(k) -> [A, l, p] (or @(k, x) -> [J, w, p] with
                  data.x0 for Gauss-Newton), data.numUnknowns, data.numBatches;
                  optional data.priors
WHAT RUNS (opts.Estimate / opts.Robust / data.x0):
  Estimate "none"    one generalised solve under the declared laws (adjustBlock)
  Estimate "noise"   the joint variance-component estimator (adjustVCEjoint):
                     one spectral shape per Pool group, one level per stream,
                     per-arc factors when ArcFactors; prior groups with
                     estimate "scale" get their sigma by Helmert / Koch-Kusche
  Robust "huber" | "igg3" | "bisquare"   IRLS on the same loop (adjustRobust)
  data.x0            Gauss-Newton with damping (adjustNonlinear)
REGULARISATION is declared, never an option: data.priors(j) on the common
parameters and data.systems(k).priors(l) on the local ones -- one struct per
group, by `group` (a name pattern) or `L` (in the columns lsqvce.layout
announces), x0, sigma, from, name, estimate. A free network's datum is the
prior on its null space.
WATCHING A RUN: Options.OnPass fires after every pass with the pass state,
the live accumulator, `data` and every system's processingOptions.

INPUT
  data   1x1 struct    the declaration (kind A, B or C as above)
  opts   1x1 lsqvce.Options   default lsqvce.Options()
  Name, Value          any property of lsqvce.Options
OUTPUT
  R      1x1 lsqvce.Results   x, sigma0, sol (Cxx, dof, vtv), diag, acc, history,
                              noise (C, Cgroups, stream, model, priors), stopReason,
                              meta (options as resolved, warnings, seeds, version);
                              methods report, summary, convergence, residuals,
                              contribution, compare, plot, toTable, coefficients

ERRORS
  Your declaration is wrong (raised BEFORE any data is read):
    lsqvce:fit:unknownField    a field name the contract does not know
                               (names the field; typos are errors on
                               purpose -- and so are the retired fields:
                               common, numCommon, numLocal,
                               observationTypes, systemId, cache, loader,
                               plotPostSolve, plotPostModel, since v4
                               localsPerArc (declare 'arc' rows in
                               localParams instead), and since v5.3
                               epochCovariance (the day returns Cep))
    lsqvce:fit:noSystems       data.systems missing, not a struct, or empty
    lsqvce:fit:noCommon        data.commonParams missing, or not a 2-D
```

```
                              cell of template rows
    lsqvce:fit:badScope       a scope-token column 1 does not carry:
                              commonParams needs 'common'|'day' (raised
                              here), localParams needs
                              'arc'|'day'|'system' (raised by makeLoader
                              and relabelled)
    lsqvce:fit:missingField   a required system field is absent, or
                              covariance carries no observation type
    lsqvce:fit:noDays         no day list anywhere
    lsqvce:fit:badId          id is not a numeric scalar or single char
    lsqvce:fit:duplicateSystem two systems share an id (explicit or
                              index default)
    lsqvce:fit:builderKind    buildDay is not a function handle
    lsqvce:fit:badArcs        arcs = 'fromDay' (ENGINE-tier only: it
                              reads data before the run; declare a
                              resolver @(day,obsType) instead)
    lsqvce:fit:<makeLoader id> a makeLoader construction error, relabelled
                              (e.g. paramType, badArcs, localPlan);
                              the original is attached as the cause
  Your choice is not available here:
    lsqvce:fit:notYet         Robust ~= "none" together with Model ~= "gls" --
                              (R5 M2 2b) the one-loop composition of M4;
                              before 2b this refused Robust on the streaming
                              kind altogether --
                              the streaming kind screens outliers
                              through its diagnostics (report())
    lsqvce:fit:modelKind      Model other than "gls" on kinds B / C
    lsqvce:batchSystem:weightMatrix  kind B/C returned a full weight matrix
                              (R5 M2: the block path takes diagonal weights;
                              a full P is a covariance law of the streaming
                              kind) -- raised by lsqvce.engine.batchSystem
    lsqvce:fit:priorNotActive  (warning, R5-D8) a prior with from > 1 in a
                              single-pass run is not applied
    lsqvce:fit:badPrior       (R5 M5 5b) data.priors is malformed (unknown
                              field, L not p x nc, sigma not a positive scalar,
                              x0 of the wrong length)
    lsqvce:fit:badEstimate    (R5 M3 3d) covariance.<type>.estimate is not
                              "none" | "scale" | "law"
    lsqvce:fit:lawOnFixedStructure  (R5 M4 4b) estimate "law" on a stream whose
                              sigma is per-row (a fixed structure: level only)
    lsqvce:fit:badMatrix      kind B shapes disagree (A m x u, l m x 1,
                              p m x 1)
    lsqvce:fit:missingField   kind C lacks batchFcn / numBatches, or the
                              linear form lacks numUnknowns
    lsqvce:fit:uCount         kind C carries BOTH x0 and numUnknowns --
                              u is stated once (numel(x0))
  Your buildDay is wrong (raised WHILE READING, by the engine, ids
  unchanged): lsqvce:makeLoader:badDay / dayShape / badArc / badArcId /
  buildDay / badDayCep / axisConflict / cepTwice (v5.3: the day's Cep and
  axis count), lsqvce:covModel:cepOnVar, lsqvce:accum:arcBadNc / arcBadNl;
  lsqvce:makeLoader:noCep is a once-per-run warning (a multi-axis day
  without Cep is composed with D = I)

See also LSQVCE.OPTIONS, LSQVCE.RESULTS, LSQVCE.LAYOUT, LSQVCE.ENGINE.DECLARATION,
     LSQVCE.ENGINE.MAKELOADER, LSQVCE.ENGINE.ADJUSTBLOCK, LSQVCE.ENGINE.ADJUSTVCEJOINT,
```

```
          LSQVCE.ENGINE.ADJUSTROBUST, LSQVCE.ENGINE.ADJUSTNONLINEAR

  Created 2026-08-31 (CEST, UTC+02:00) with assistance from Claude (Anthropic),
  model Claude Fable 5 -- R4 Level L2, WP-56 (decision R4-D4).
  Help rewritten 2026-09-04 for v5.2 (R5 M5 / R5-D8), model Claude Fable 5.1:
  the v5 surface (Estimate, priors, paramColumns, OnPass, layout, toTable).
  Updated 2026-09-06 (CEST, UTC+02:00), R6 WP-90: the error information
  travels with the day (D.Cep); epochCovariance retired; the See also on
  the units that exist. With assistance from Claude (Anthropic), model
  Claude Fable 5.1.
```

## 5.2 lsqvce.Options

LSQVCE.OPTIONS The one settings object for LSQVCE.FIT (v5: 14 knobs + 5 groups).

```
LSQVCE.OPTIONS  The one settings object for LSQVCE.FIT (v5: 14 knobs + 5 groups).

  opts = lsqvce.Options()                        every default
  opts = lsqvce.Options(Name, Value, ...)        override some
  opts.Name = Value                              change one later (validated)
  R = lsqvce.fit(data, Name, Value, ...)         the same names, straight on fit

  WHAT TO ESTIMATE, HOW TO RUN. Everything ABOUT A STREAM (its law, whether
  its law / level / arc factors are estimated) is declared on the stream
  (data.systems(k).covariance.<type>: type, sigma, estimate, arcFactors);
  this object holds only RUN policy. R5 M5 (WP-87, decision R5-D2) cut it
  from 62 flat knobs to the 14 below plus five typed groups; the migration
  from every v4 name is in RELEASE_NOTES.md and HANDOVER §12.10.

  TOP LEVEL
    Estimate   "none" | "noise"   "none": one GLS solve under the declared
                                  laws. "noise": the joint (Ellmer) variance
                                  component estimator on every stream whose
                                  spec allows it (estimate ~= "none"), with
                                  the policy below. Replaces Model: the v4
                                  spellings are Estimate="noise" plus
                                    "ellmer" -> Pool as set, ArcFactors true
                                    "vce"    -> Pool "stream", ArcFactors false
                                    "pooled" -> Pool "all",    ArcFactors false
    Pool       "all" | "obsType" | "stream"   which streams SHARE one estimated
                                  spectral shape (every stream keeps its own
                                  level regardless); default "all"
    ArcFactors logical            per-arc variance factors on (Ellmer 6.5.4),
                                  default true; false = one level per stream.
                                  Only the RATIO between a stream's arcs is
                                  identifiable -- the level absorbs their mean
                                  (R6 F-01, fixed in WP-98: the ratio settles
                                  across passes, 9.4 / 8.6 / 8.6 / 8.6 against
                                  a planted 9; before it grew 8.4x a pass).
    Robust     "none" | "huber" | "igg3" | "bisquare"   IRLS on the same
                                  loop, every kind (details in RobustOpts)
    MaxPasses  double  10         the ONE pass budget (v4: MaxOuter / JointMaxIter)
    MinPasses  double  1          passes run even if the tolerance is met
                                  (GRAVFIRE's minimumIterations)
    Tolerance  double  1e-2       stop when max|alpha2 - 1| over frequencies
                                  and arcs is below it (v4: JointTol)
```

```
  Backend    "qr" | "normal"    square-root vs normal equations
  Probes     double  100        Hutchinson probes for the per-lag
                                redundancies (0 = the exact route)
  Seed       [] | double        RNG seed for the probes (recorded)
  CacheDays  double  1          built days kept resident per system; Inf =
                                the whole calendar (one build per run)
  Verbose    logical | 0..2     screen messages (0 silent, 1 progress,
                                2 detail; level 3 was merged into 2 in
                                2026-07 and lsqvce.util.verbosity refuses it)
  LogFile    string  ""         mirror the messages to this file ("" = none)
  OnPass     [] | @(ctx)        YOUR inspection hook, called at the end of
                                EVERY pass of every iterative run (the
                                variance-component loop, IRLS, Gauss-Newton)
                                with ONE ctx struct: .driver .pass .x .Cxx
                                .sigma0, the loop state (noise: .C .Cgroups
                                .sigma2 .arcKeys .streams .sigma .white
                                .whitePval .maxDev .skips; IRLS: .factor
                                .keys .scale .dx; Gauss-Newton: .vPv .dxRel
                                .lambda), .acc (the LIVE accumulator: call
                                lsqvce.engine.residuals(ctx.acc, ...) or any
                                engine function inside the hook), .data (your
                                declaration as passed to fit) and
                                .processingOptions (per system, as declared).
                                Plot, drawnow, save wherever you like; return
                                nothing to continue, false to stop after this
                                pass (stopReason "userStop"). A throwing hook
                                is NOT swallowed (lsqvce:callUser:threw with
                                your error as cause). Replaces v4's
                                plotPostModel / plotPostSolve seams.

GROUPS (structs; set fields one at a time or give a whole struct)
  Noise        Grid (60 | "auto")  frequencies of the estimated shape
               Whitener "toeplitz"|"ar"|"arma"  how a PSD law whitens (v4 PsdMode)
               MeanType "robust"|"plain"        arc-factor normalisation
               Smooth 0                         moving average on the PSD
               FreezeRedundancy 1               components below it hold
               SkipThreshold 0.2                skip RATE that refuses the pass
  RobustOpts   Tuning [], MaxIter 30, Tol 1e-8
  Diagnostics  Alpha 1e-3, Bonferroni true, Beta 0.2, RMin 0.01, RWeak 0.1,
               Test "auto"|"pope"|"baarda"|"mad", RobustCheck true,
               Store "all"|"flagged"|"none", NWorst 1000
  Parallel     Mode "serial"|"parfor"|"spmd", DistributeDays false, Threshold 8
  GaussNewton  TolF 1e-10, Lambda0 1e-6     (x0 problems; MaxPasses / Tolerance
               are the budget and the step tolerance)

REGULARISATION is not an option: it is DECLARED on the data, as
pseudo-observations in the user's own columns -- data.priors(k) on the
common parameters (L in the columns of A) and data.systems(k).priorDay on
the local ones (L in the columns of B), each with x0, sigma, from, group
(R5-D8). The v4 Reg / RegLambda / RegMatrix / DatumB / Sigma0 / Sparse
knobs are gone (HANDOVER §12.10).

METHODS
  engineArgs(target)   the Name-Value list a driver takes (validated)
  toStruct()           every public property as a struct (provenance)
```

```
  disp()               grouped display, engine defaults marked *

ERRORS
  MATLAB:validators:*          a typed property receives a value outside
                               its type / set / range (at assignment)
  lsqvce:Options:badGrid       Noise.Grid is neither "auto" nor an integer >= 2
  lsqvce:Options:badVerbose    Verbose is neither logical nor an integer 0..2
  lsqvce:Options:badOnPass     OnPass is neither [] nor a function handle
  lsqvce:Options:badGroupField a group struct carries an unknown field
  lsqvce:Options:badTarget     engineArgs target is unknown

EXAMPLE
  opts = lsqvce.Options(Estimate="noise", Pool="obsType", Robust="huber");
  opts.Noise.Grid = 2160;  opts.Parallel.Mode = "parfor";
  R = lsqvce.fit(data, opts);

See also LSQVCE.FIT, LSQVCE.RESULTS, the Manual, chapter 3, RELEASE_NOTES.md

Updated 2026-09-08 (CEST, UTC+02:00), R6 WP-98: ArcFactors says what is
identifiable (the ratio) now that F-01 is fixed. With assistance from
Claude (Anthropic), model Claude Opus 5.
Created 2026-09-03 (CEST, UTC+02:00) with assistance from Claude
(Anthropic), model Claude Fable 5.1 -- R5 Level M5, WP-87 (v5 interface).
```

## 5.3 lsqvce.Results

LSQVCE.RESULTS The outcome of a recovery: solution, diagnostics, provenance.

```
LSQVCE.RESULTS  The outcome of a recovery: solution, diagnostics, provenance.
  R = lsqvce.Results(sol, D, acc, diagSol, batchFcn, nBatch, systems)
  R = lsqvce.Results(sol, D, acc, diagSol, batchFcn, nBatch, systems, meta)

 Returned by LSQVCE.ENGINE.ADJUSTBLOCK (and, later, LSQVCE.ENGINE.ADJUSTVCE). It gathers
  everything a user needs to read and judge an adjustment in one object, and
  keeps the lower layers reachable for experts. You rarely construct it
  yourself; you receive it from adjustBlock and call R.report().

  PROPERTIES (read-only)
    x          nc x 1   double   Common-parameter estimate (the field).
    sigma0     1x1      double   A-posteriori variance factor (global).
    stopReason 1x1      string   Why the loop stopped (R5 M1): "single" |
                                 "converged" | "maxPasses". Convergence
                                 failure is a RESULT: the last pass is
                                 returned with its covariance and one warning.
    converged  1x1      logical  Did the pipeline converge? (true for a single
                                 solve; set by the iterative engines.)
    diag       1x1      struct   Per-observation diagnostics (LSQVCE.ENGINE.DIAGNOSE):
                                 redundancy, leverage, w-tests, MDB, ...
    history    1xI      struct   Per-iteration record (empty for a single
                                 solve; filled by LSQVCE.ENGINE.ADJUSTVCE).
    noise      1x1      struct   Fitted noise models per stream (empty until
                                 the separable/VCE loop populates it).
    sol        1x1      struct   Raw output of acc.solve() (.x, .sigma0, ...).
    acc        1x1      lsqvce.engine.BlockAccumulator   The accumulator (drop here
                                 to re-solve, inspect blocks, get residuals).
    source     1x1      struct   Bridged solution from LSQVCE.ENGINE.DIAGNOSEACCUM.
```

```
  batchFcn   1x1      function_handle   Re-reader used by LSQVCE.ENGINE.DIAGNOSE.
  nBatch     1x1      double   Batch count from the bridge.
  systems    1xM      struct   The resolved system specs (sysID/nl/dayList).
  meta       1x1      struct   Run provenance (WP-44). Field list is
                               ADD-ONLY across versions, never renamed:
                               .toolboxVersion 1xK char  from Contents.m,
                                               the ONE version source
                               .matlabRelease  1xK char  version('-release')
                               .createdUTC     1xK char  ISO 8601, trailing Z
                               .seeds          1x1 struct every RNG seed the
                                               run resolved (struct() when
                                               none; a resolved-but-unset
                                               seed is stored as [])
                               .options        1x1 struct the RESOLVED
                                               options the run actually used
                              A bare construction (7 arguments) gets a
                              meta with empty seeds/options -- the field
                              is never missing.

METHODS
  V = R.report(Name,Value,...)   Print/return the level-0..4 verdict
                                 (LSQVCE.REPORT); options pass straight through.
  r = R.redundancy()             Total redundancy sum(r_i) = n - u.
  res = R.residuals(Name,Value)  Whitened/raw residuals (acc.residuals).
  con = R.contribution(...)      Per-group information share.
  T = R.convergence(Name,Value)  Per-ITERATION trajectory: when the loop
                                 settled (needs .history).
  F = R.compare(Rred)            Nested-model F test against a reduced fit (R5 M7).
  fh = R.plot(kind)             "residuals" | "contribution" | "psd" | "convergence"
                                 (R5 M5: one figure per question, nothing exported).
  [C, L, map] = R.toTable(data)  The expanded parameter tables with estimate and
                                 standard deviation written in (R5-D8).
  T = R.summary()                Per-STREAM final state: sigma, AR order,
                                 whiteness (needs .history).
  T = R.coefficients(Alpha=a)    The estimates as a table with formal
                                 errors and (1-a) confidence bounds
                                 (R4 WP-57; default a = 0.05).
  disp(R)                        One-screen summary.

THE TWO ITERATION READERS ARE COMPLEMENTARY. convergence() reduces across
STREAMS (one row per pass); summary() reduces across PASSES (one row per
stream). A converged loop can still hide a single stream that never
whitened -- the per-pass view reports the worst p-value without naming it.

EXAMPLE
  R = lsqvce.engine.adjustBlock(systems, days);
  R.report('Level', 2);
  if ~R.converged, lsqvce.util.log.warn('did not converge'); end
  x = R.x;   s0 = R.sigma0;

ERRORS
  lsqvce:Results:noVPv          compare() on a Results without a finite v'Pv
  lsqvce:Results:noHistory      an iteration reader was called on a
                                single-solve Results (adjustBlock does not
                                iterate); use the result of adjustVCE
```

```
  lsqvce:Results:noCovariance   coefficients() on a Results whose sol
                                carries no Cxx
  lsqvce:Results:noAccumulator  residuals()/contribution() on a Results
                                without an accumulator (fit kinds B / C)

ACCURACY
  Results adds no numerics: x and sigma0 are the engine's (sigma0 is the
  A-POSTERIORI std of unit weight; sol.Cxx the covariance scaled by it --
  the streaming contract every producer must establish, R4 §11.8 L4 r1),
  report() is lsqvce.report on .diag, coefficients() is arithmetic on x
  and diag(Cxx) with z(0.05) = 1.959963984540054 in closed form.

See also LSQVCE.FIT, LSQVCE.ENGINE.ADJUSTBLOCK, LSQVCE.ENGINE.ADJUSTVCE, LSQVCE.REPORT,
         LSQVCE.ENGINE.DIAGNOSE, LSQVCE.ENGINE.BLOCKACCUMULATOR

Updated 2026-09-01 (CEST, UTC+02:00), R4 WP-61 (Level L3): ACCURACY block
(the sigma0 / Cxx contract stated once). With assistance from Claude
(Anthropic), model Claude Fable 5.
Updated 2026-08-31 (CEST, UTC+02:00), R4 WP-56b (Level L4): residuals()
and contribution() refuse with lsqvce:Results:noAccumulator on a Results
without an accumulator (the in-memory / batched kinds of fit). With
assistance from Claude (Anthropic), model Claude Fable 5.
Updated 2026-08-31 (CEST, UTC+02:00), R4 WP-57 (Level L2): coefficients()
-- estimates, formal errors and confidence bounds as ONE table (the
fitlm Coefficients/coefCI pair folded into one method; z from erfcinv,
no Statistics Toolbox). With assistance from Claude (Anthropic), model
Claude Fable 5.
Updated 2026-08-19 (CEST, UTC+02:00), WP-44 (T-05): run provenance --
immutable .meta composed by the ONE private builder runMeta, constructor
widened to an optional 8th argument, report() forwards the provenance
line. With assistance from Claude (Anthropic), model Claude Fable 5.
Updated 2026-08-19 (CEST, UTC+02:00), WP-39 stage (c) c2 (F-07):
typed `arguments` signatures, canonical UpperCamelCase option names
(legacy casings keep matching -- MATLAB NV semantics; enum VALUES
case-sensitive lowercase). With assistance from Claude (Anthropic),
model Claude Fable 5.
Created 2026-07-16 12:20 (CEST, UTC+02:00) with assistance from
Claude (Anthropic), model Claude Opus 4.8.
```

## 5.4 lsqvce.report

LSQVCE.REPORT Judge a least-squares adjustment and say so in words.

```
LSQVCE.REPORT  Judge a least-squares adjustment and say so in words.

  V = LSQVCE.REPORT(D) takes the struct from lsqvce.engine.diagnose and prints a
  verdict. It returns a struct of machine-readable findings so the same logic
  can drive an automated pipeline.

  The report is ordered so that a failure at one level invalidates the levels
  below it. It refuses to report parameter precision for an adjustment whose
  integrity checks failed, because a sigma computed from a broken system is
  not a conservative estimate -- it is a fabrication.

  LEVELS
    0  INTEGRITY   sum(r) == n-u, vPv >= 0, vPv consistent between passes.
```

```
                These test the CODE, not the data. Failure here means
                every number below is meaningless.
  1  GLOBAL      variance factor vs chi-square.
  2  LOCALISE    outliers (data snooping), with Bonferroni.
  3  RELIABILITY redundancy numbers, MDB, external reliability.
                This is the level that says which observations you are
                NOT ABLE TO CHECK.
  4  PARAMETERS  correlations, conditioning, precision.

NAME-VALUE
  'verbose'  true  print the report (default true)
  'width'    78    line width
  'Provenance' struct()  a run-provenance struct (LSQVCE.RESULTS .meta,
                  WP-44); when non-empty, ONE level-0 NOTE line
                  (code 'PROVENANCE') states toolbox version, MATLAB
                  release, and creation time. R.report() forwards it
                  automatically; a bare lsqvce.report(D) prints none.

OUTPUT V
  V.verdict   'PASS' | 'PASS WITH WARNINGS' | 'SUSPECT' | 'INVALID'
  V.findings  cell array of struct(level, severity, code, msg)
  V.nError, V.nWarn, V.nNote

INPUT
  D         1x1 struct  diagnostics from LSQVCE.ENGINE.DIAGNOSE
  varargin  Name/Value: 'verbose', 'width', 'alpha', 'rWeak', ...

EXAMPLE
    data = struct('batchFcn', src, 'numUnknowns', u, 'numBatches', 4);
    R = lsqvce.fit(data, Verbose=false);
    V = lsqvce.report(R.diag);         % prints the five levels (= R.report)

    % V.verdict    'PASS' | 'PASS WITH WARNINGS' | 'SUSPECT' | 'INVALID'
    % V.findings   machine-readable: {level, severity, code, msg}

    % Drive a pipeline off the codes rather than the text:
    codes = cellfun(@(f) f.code, V.findings, 'UniformOutput', false);
    if ismember('SNOOP_CONTAMINATED', codes)
        error('Pope is blind here -- re-run with test=mad.');
    end

Updated 2026-08-19 (CEST, UTC+02:00), WP-39 stage (b) b2 (F-07):
typed `arguments` signature, canonical UpperCamelCase option names
(legacy casings keep matching -- MATLAB NV semantics; enum VALUES
case-sensitive lowercase). With assistance from Claude (Anthropic),
model Claude Fable 5.
See also LSQVCE.ENGINE.DIAGNOSE, LSQVCE.ENGINE.SYSTEMATIC.

Generated with assistance from Claude (Anthropic), Claude Opus 4.8,
2026-07-11.
```

## 5.5 lsqvce.layout

LSQVCE.LAYOUT The column layout fit will use – so you can declare priors in it.

```
LSQVCE.LAYOUT  The column layout fit will use -- so you can declare priors in it.
```

```
lay = lsqvce.layout(data)

The toolbox derives from your parameter tables (scope tokens, calendar,
arcs, validity) which column of A and of every system's B belongs to
which instantiated parameter. This function announces that layout BEFORE
fit runs, so that data.priors(j).L (columns of A) and
data.systems(k).priors(l).L (columns of B) can be written in it -- the
only knowledge the user's regularisation and the toolbox share (R5-D8).

INPUT   data  1x1 struct  a kind-A declaration (the Manual, chapter 3)
OUTPUT  lay   1x1 struct
  .nc            1x1        columns of A
 .common     nc x 1 struct (.row .day)    the template row and (for 'day' rows) the day of every col
  .systems(k)    1x1 struct per system:
     .nl         1x1        columns of B
   .local    nl x 1 struct (.row .day .arc) the template row / day / arc of every column of B
    .col      @(row, day, arc) -> column    lookup: the column of template row `row` on
                                       `day` (and `arc` for 'arc' rows; [] otherwise);
                                            0 when that instance does not exist
 .colCommon    @(row, day) -> column        the same for A ([] day for 'common' rows)

EXAMPLE  (a Tikhonov prior on local row 7 of system 1, on every day)
  lay = lsqvce.layout(data);
  nl  = lay.systems(1).nl;  rows = zeros(0, nl);
  for d = data.days
    c = lay.systems(1).col(7, d, []);  r = zeros(1, nl);  r(c) = 1;  rows(end+1, :) = r;
  end
 data.systems(1).priors = struct('L', rows, 'x0', 0, 'sigma', 0.01, 'name', 'xaccscale', 'estimate

See also LSQVCE.FIT, LSQVCE.RESULTS/TOTABLE, the Manual, chapter 3

Created 2026-09-03 (CEST, UTC+02:00) with assistance from Claude
(Anthropic), model Claude Fable 5.1 -- R5-D8 round 5.
```

# 6 Regularisation: priors and groups

## 6.1 The rule

*Parameters and their columns are yours* — the tables, `buildDay`, plots, `toTable`. *Regularisation is the toolbox's* — the mechanism, the redundancy, the activation pass, the estimation of its strength. The interface between them is an operator in your columns, announced by `lsqvce.layout`, checked by width.

## 6.2 What a prior is

Tikhonov regularisation of a parameter is an additional observation equation, "parameter j is observed to be $x_0$ with standard deviation σ":

$$L \cdot x = x_0 + v, \quad \text{weight } 1/\sigma^2$$

where L is the coefficient matrix of the pseudo-observations (one row per pseudo-observation) exactly as A is the coefficient matrix of the observations. The toolbox therefore treats a prior as data: it enters the normal equations, it has a redundancy (how much of it the data overrule), and estimating its σ is variance-component estimation on that group (Koch–Kusche).

## 6.3 The declaration — one mechanism for common and local parameters

```
data.priors(j)            = struct(...)    % on the common parameters, L in the columns of A
data.systems(k).priors(l) = struct(...)    % on system k's local parameters, L in the columns of
```

One struct per **group**, with the fields

| field | meaning |
|---|---|
| `group` | a regular expression matched against the *name column* of the table (column 2; `data.paramColumns.name` to say otherwise). The toolbox builds L: one row per matched template row per instance (per day for a `'day'` row, per arc for an `'arc'` row, once for `'system'`/`'common'`). |
| `L` | the explicit operator (q × nc or q × nl) in the columns that `lsqvce.layout(data)` announces. Give `group` **or** `L`, never both. |
| `x0` | the a-priori values (q × 1, or a scalar; default 0) |
| `sigma` | ONE standard deviation for the rows of the group. Relative weights inside a group go into L (divide a row by its own relative σ); the struct's σ is the common factor. |
| `from` | the pass at which the group becomes active (default 1). In a single solve a `from > 1` group is reported (`lsqvce:fit:priorNotActive`) and skipped. |
| `name` | the label (default the pattern); local groups become the stream `prior_<name>`. |
| `estimate` | `"none"` (default; σ fixed) or `"scale"`: σ is estimated every pass of `Estimate="noise"` by the Helmert rule $\sigma^2 \leftarrow \sigma^2 \cdot \Omega/r$ — the fixed point $\Omega = r$ ("the prior's weighted residual sum equals its redundancy"), so $\lambda = s_0^2/\sigma^2$ is the regularisation the data support. |

**Several rows make σ estimable.** A group with one row has redundancy ≤ 1 and cannot estimate its own σ (it is left unchanged and flagged in `R.noise.priorHistory`). The x/y/z accelerometer scales of both satellites over a month are one group with 180 rows.

**Local rows live in one day.** A row of a local group has its non-zero columns inside one day's local block (or the system block); coupling two days is done through the common parameters. Different days of the same parameter are different rows — that is how the estimate pools information across the calendar.

## 6.4 How they are treated

Common groups are stacked into one prior on the common block, folded into the solve and appended to the diagnostic sweep (they carry a redundancy). Local groups ride as their own observation stream `prior_<name>` of the system — pseudo-observation arcs with A = 0, B = L/σ, l = $x_0/\sigma$ and a unit law — so accumulation, elimination, the sweeps, `residuals()` and the redundancy count see them as data, and under `Estimate="noise"` the group's σ is the stream's level: identical behaviour to the common groups, no second estimator. `R.noise.priors`, `R.noise.priorHistory` (common) and `R.noise.model{k}.sigma` for stream `satKey|prior_<name>` (local) report the result.

### 6.5 The datum of a free network

Minimum constraints $B^T x = 0$ are the group `L = B', x0 = 0`: when B spans the null space the solution is the exact minimum-constraint one for **any** weight (the data's range and the null space are orthogonal). Without it the accumulator refuses a rank-deficient system by name (`lsqvce:qr:singularR`); it never guesses.

### 6.6 The layout

```
lay = lsqvce.layout(data);
lay.nc, lay.common(j)                    % the template row (and day) of every column of A
lay.systems(k).nl, lay.systems(k).local(c)   % (row, day, arc) of every column of B
c = lay.systems(k).col(row, day, arc); c = lay.colCommon(row, day);   % lookups (0 = not instanti
```

### 6.7 Your table's column 11

`{'Tikhonov', fromPass, sigma[, label]}` on a row means exactly one of the structs above; `template/gravfire_data.m` (STEP 6/7) emits them: rows sharing a label become one group in the `group` form with `estimate="scale"`, rows without a label one fixed-σ group each.

## 7 Stochastic models and their estimation

Every observation stream carries a covariance spec in `data.systems(k).covariance.<type>` (chapter 3 lists the fields) — its TEMPORAL law: `diag` (white), `arma` / `ar` (a whitening filter), `acf` / `psd` (an exact Toeplitz law). The per-epoch accuracy of the observations (a sigma per epoch, or a full d × d covariance per epoch for a multi-axis stream) is not declared: `buildDay` returns it with the day as `D.Cep` (chapter 3, §7) and the toolbox composes C = D (R_t ⊗ I_d) D per arc — for every temporal law, scalar streams included. Batches are overlapped by the filter memory automatically, so batching never changes the result.

With `Estimate="noise"` the joint estimator (theory ch. 6, Ellmer 2018) runs on every stream whose spec allows it (`estimate = "law"` default; `"scale"` = level only, the declared shape kept; `"none"` = untouched). Per pass it accumulates and solves under the current laws, sweeps every arc once for its whitened residual and Hutchinson-probed redundancies at every lag (all lags from one FFT per column), updates the spectral shape per pool group in the PSD domain (positive by construction, resampled to every arc's length), the level per stream, the per-arc factors, and the prior groups' σ; it stops at `Tolerance` or `MaxPasses`. A law with a per-row sigma is a fixed structure: it can carry a level but not a spectral shape.

What comes back: `R.noise.Cgroups` (one covariance function per pool group), `R.noise.stream` / `R.noise.model{k}` (per stream: the level, its law, its arc factors), `R.history.streams / sigma / white / whitePval` per pass (the Ljung–Box whiteness of every stream, K = 20, α = 0.05), `R.convergence()`, `R.plot("psd")`, `R.plot("convergence")`.

The acceptance of the estimator is physical: a planted field and planted stream sigmas are recovered within their formal errors (`demo_gravity`, counted in the test suite).

## 8 Diagnostics and the verdict

Every solve carries per-observation diagnostics in `R.diag` (residuals in the whitened frame `sqrt(p).*v`, redundancy numbers, standardised residuals, the Baarda / Pope test with `Diagnostics.Alpha` and Bonferroni, minimal detectable biases, reliability, the a-priori backstop). `R.report(Level=0..4)` reduces them to one verdict and its evidence; `R.summary()` and `R.convergence()` read the pass record of an iterative run; `R.residuals()` re-reads the arcs for the whitened residuals per stream; `R.contribution()` gives the information share of every

stream in the common parameters; `lsqvce.engine.snoop` iterates data snooping with a clean re-solve on the block path; `R.compare(Rred)` is the nested-model F test between two fits. For correlated batches the per-observation quantities live in the whitened frame; only their sums are frame-invariant.

# 9 Watching a run

`Options.OnPass` (chapter 4) is called after every pass with the pass state, the live accumulator, your declaration and your `processingOptions` — plot, `drawnow`, save, or return `false` to stop. Afterwards `R.plot("residuals" | "contribution" | "psd" | "convergence", Stream=…, Visible=…)` returns one figure per question; nothing is exported by the toolbox. `R.meta.warnings` lists every warning identifier of the run; `R.stopReason` says why the loop stopped (`"single"`, `"converged"`, `"maxPasses"`, `"userStop"`).

```
function go = myInspect(ctx)
    res = lsqvce.engine.residuals(ctx.acc, struct('x', ctx.x, 'sigma0', ctx.sigma0, ...
                                                  's0sq', ctx.s0sq, 'Cxx', ctx.Cxx), 'Verbose', f
    figure(1); plot(res.sys(1).obs(1).t, res.sys(1).obs(1).w, '.'); drawnow;
    go = ctx.maxDev > 1e-3;                % or return nothing to continue
end
```

# 10 The engine, kernel and util tiers

What `fit` dispatches to and what it is built from. One line per public unit, from the `Contents.m` files (generated). `doc lsqvce.engine.<name>` opens the full page; the tier rule (HANDOVER, tiers.csv): the user tier calls engine, engine calls kernel and util, never upward.

## 10.1 lsqvce.engine — drivers, accumulator, loaders, sweeps, diagnostics

| unit | what |
|---|---|
| `adjustNonlinear` | Gauss-Newton / Levenberg as a policy on the block path (R5 M2) |
| `batchSystem` | a batch source as one local-free streaming system (R5 M2) |
| `adjustRobust` | IRLS (Huber/IGG3/bisquare) as a policy on the block path (R5 M2) |
| `adjustBlock` | one GLS solve over the streaming block-arrow pipeline |
| `adjustVCEjoint` | Ellmer ten-step joint co-estimation with keyed arc factors |
| `applyRegularisation` | regularisation of the reduced system |
| `arcVarianceFactors` | per-arc variance factors |
| `BlockAccumulator` | block-arrow accumulation with per-system Schur pre-elimination |
| `checkPSD` | PSD sanity check |
| `collectResiduals` | residual collection per stream |
| `CommonPrior` | prior on the common block (regularisation as observations) |
| `contribution` | information contribution per group |
| `covComponent` | covariance component estimation |
| `covModel` | covariance spec -> whitening operators (per arc) |

| unit | what |
|---|---|
| `datum` | datum defect detection and handling |
| `declaration` | validate a kind-A data struct and build its loaders (R5-D8) |
| `diagnose` | per-observation diagnostics (batch) |
| `diagnoseAccum` | diagnostics bridge from the accumulator |
| `ellipse` | confidence ellipse from a covariance |
| `ftest` | F-test on variance factors |
| `gcvLambda` | GCV choice of the regularisation weight |
| `jointModelOpt` | joint-mode model -> loader UserOpt translation |
| `leverage` | leverage (hat-matrix diagonal), batch |
| `leverageBlocks` | leverage over blocks |
| `localDiagnostics` | diagnostics of the local blocks |
| `localColumns` | which (row, day, arc) owns each column of B (R5-D8) |
| `makeLoader` | day-builder -> Loader (built INSIDE fit from the data struct, R4-D4) |
| `mapreduce` | serial / parfor / spmd task runner with ordered reduce |
| `NormalEqSystem` | normal-equation backend |
| `outliers` | outlier listing from diagnostics |
| `psdVCE` | PSD amplitude + arc-factor VCE (Ellmer workhorse) |
| `QRSystem` | QR backend |
| `ReducedContribution` | reduced-normal contribution bookkeeping |
| `redundancy` | redundancy numbers (data pass) |
| `residuals` | second data pass: residuals and quadratic forms |
| `resolveNc` | common-parameter count derived from the systems |
| `satKeyOf` | satellite key (makeValidName) helper |
| `synthLoader` | synthetic block-arrow loader (engine-tier fixture; leaves in M4) |
| `sharedProbe` | shared Hutchinson probe generator (RNG contract) |
| `snoop` | block-native data snooping and downweighting |
| `stat` | statistics helper |
| `streamContribution` | contribution per stream |
| `systematic` | systematic-effect test |
| `traceEstimate` | stochastic trace estimate |
| `vce` | Helmert group variance components (exact |
| `runMeta` | immutable provenance record for Results.meta |

## 10.2 lsqvce.kernel — the numerical kernels

| unit | what |
|---|---|
| `ar2arma` | AR -> ARMA conversion |
| `armaACV` | ARMA autocovariance |
| `armaWhiten` | ARMA whitening filter (batched) |
| `crossCorr` | cross-correlation of residual streams |
| `dctI` | DCT-I transform |
| `decorrFactor` | decorrelation factor of a dense covariance |

| unit | what |
|---|---|
| `levinsonDurbin` | Levinson-Durbin recursion |
| `psd2acv` | PSD -> autocovariance |
| `redundancyLags` | redundancy per lag without the dense W^-1 |
| `separableWhiten` | separable (temporal x axis) whitening |
| `toeplitzCofactor` | Toeplitz cofactor construction |
| `varACV` | VAR autocovariance |
| `varWhiten` | VAR whitening |
| `whitenessCorr` | whiteness (Ljung-Box) statistic |
| `whittle` | multichannel Levinson (Whittle) VAR fit (moved from engine, R5 M2) |
| `robustWeights` | IRLS weight factors from standardised residuals (R5 M2) |

### 10.3 lsqvce.util — plumbing

| unit | what |
|---|---|
| `callUser` | call a user function: outputs counted once, errors wrapped (R5 M1) |
| `getfielddef` | struct field with default |
| `toolboxVersion` | the version string from Contents.m (one source) (v5.2) |
| `onPass` | fire the per-pass inspection hook (Options.OnPass) (R5 M5) |
| `rowWeightKey` | the one key convention for per-arc robust weights (R5 M2) |
| `log` | toolbox-wide logging state |
| `mergeStruct` | struct merge (later wins) |
| `verbosity` | verbosity level resolution and formatting |

## 11 Theory

`lsqvce_theory.pdf` (the R5 theory, 54 pages) is the authority for every formula the toolbox implements: the block-arrow adjustment and its two backends, the joint variance-component estimator (the per-lag redundancies, the PSD-domain law, the per-stream level and the arc factors), priors and Koch–Kusche, the diagnostics (Baarda, Pope, reliability, whiteness), IRLS and Gauss–Newton, and Appendix C, the option mapping as implemented. The literature: Ellmer (2018), Kvas (2019), Koch and Kusche (2002), Baarda (1968), Pope (1976).

## 12 Migration from 4.x to 5

Every 4.x declaration still fits; three things were added (priors, per-stream policy, `OnPass`), the option space was cut to 14 knobs + 5 groups, and every retired name has the successor below. Numbers may differ from 4.0 because the separate-mode estimator (an AR law fitted to a biased raw-residual ACF) is gone; the acceptance of its successor is the physical oracle, not equality to 4.0.

### 12.1 The migration table (from RELEASE_NOTES.md)

| v4 | v5 |
|---|---|
| `Model="gls"` | `Estimate="none"` (default) |
| `Model="ellmer"` | `Estimate="noise"` (+ `Pool`, `ArcFactors=true` default) |
| `Model="vce"` | `Estimate="noise"`, `Pool="stream"`, `ArcFactors=false` |
| `Model="pooled"` | `Estimate="noise"`, `Pool="all"`, `ArcFactors=false` |
| `Robust="irls"` + `Influence` | `Robust="huber"\|"igg3"\|"bisquare"` |
| `Tuning`, `RobustMaxIter`, `RobustTol` | `RobustOpts.Tuning` / `.MaxIter` / `.Tol` |
| `MaxOuter`, `JointMaxIter`, `VceMaxIter`, `NonlinearMaxIter` | `MaxPasses` |
| `JointTol`, `OuterTol`, `VceTol`, `TolX` | `Tolerance` |
| `TolF`, `Lambda0` | `GaussNewton.TolF` / `.Lambda0` |
| `NumLags` | `Noise.Grid` |
| `PsdMode` | `Noise.Whitener` (`"toeplitz"` default, as before) |
| `MeanType`, `Smooth`, `MinRedundancy`, `SkipThreshold` | `Noise.MeanType` / `.Smooth` / `.FreezeRedundancy` / `.SkipThreshold` |
| `Parallel`, `DistributeDays`, `ParallelThreshold` | `Parallel.Mode` / `.DistributeDays` / `.Threshold` |
| `Alpha`, `Bonferroni`, `Beta`, `RMin`, `RWeak`, `Test`, `RobustCheck`, `Store`, `NWorst` | `Diagnostics.<same>` |
| `LogFid`, `LogFlag`, `ShowSeconds`, `ExportDir`, `plotPostSolve`, `plotPostModel` | `LogFile`; `OnPass`; `R.plot(kind)` |
| `DatumB` | `data.priors = struct('L', B., 'sigma', 1, 'name', 'datum')` |
| `Sigma0`, `Sparse`, `Verify`, `Progress` | retired |
| `GroupBy`, `ComponentsPerEpoch`, `MaxOrder`, `AcvTol`, `MaxLag`, `WhiteAlpha` | retired with the separate mode (groups are streams; the axis count comes from the builder; whiteness is fixed at K = 20, α = 0.05) |
| `Norm`, `C0`, `StepSchedule`, `AvgTail` | engine-tier only (`lsqvce.engine.adjustVCEjoint`) |
| `lsqvce.engine.adjust` / `robust` / `solveNonlinear` / `Accumulator` / `adjustVCE` / `Combine` / `BatchIndex` / `varComp` / `vceHutchinson` | gone; `fit` on the block path; `lsqvce.engine.adjustBlock` / `adjustRobust` / `adjustNonlinear` / `adjustVCEjoint` / `batchSystem` at engine tier |
| `R.source.verdict`, `history.nOuter` | `R.report().verdict`, `history.nIter` |

Full ledger, decisions and round notes: `dev/HANDOVER.md` §12; theory: `dev/lsqvce_theory_r5.pdf`.

# 13 Appendix A — error and warning identifiers

Every identifier the package raises, from `dev/docs_src/error_ids.csv` (the catalogue the audit keeps closed both ways: nothing raised that is not listed, nothing listed that is not raised).

| identifier | where / what |
|---|---|
| `lsqvce:Options:badGrid` | Options — error — v4.0 |
| `lsqvce:Options:badGroupField` | Options — error — v4.0 |
| `lsqvce:Options:badOnPass` | Options — error — v4.0 |
| `lsqvce:Options:badTarget` | Options — error — v4.0 |
| `lsqvce:Options:badVerbose` | Options — error — v4.0 |

| identifier | where / what |
|---|---|
| lsqvce:Results:badPlotKind | Results — error — v4.0 |
| lsqvce:Results:noAccumulator | Results — error — v4.0 |
| lsqvce:Results:noCovariance | Results — error — v4.0 |
| lsqvce:Results:noHistory | Results — error — v4.0 |
| lsqvce:Results:noVPv | Results — error — v4.0 |
| lsqvce:Results:tableMismatch | Results — error — v4.0 |
| lsqvce:accum:arcBadNc | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:arcBadNl | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:arcNotFinite | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:arcRowMismatch | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badArc | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badCxx | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badLoader | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badMeta | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badPrior | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badSpecs | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badThresholds | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:badX | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:blockSchurFailed | engine.BlockAccumulator — warning — v4.0 |
| lsqvce:accum:cancellation | engine.BlockAccumulator — warning — v4.0 |
| lsqvce:accum:duplicateSystem | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:localNotRedundant | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:localSummarySkipped | engine.BlockAccumulator — warning — v4.0 |
| lsqvce:accum:metaMismatch | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:noArcs | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:noDays | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:noStore | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:noSystems | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:notWholeEpochs | engine.private.arcBatchWhiten — error — v4.0 |
| lsqvce:accum:pinvFallback | engine.BlockAccumulator — warning — v4.0 |
| lsqvce:accum:priorNotPD | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:priorSize | engine.BlockAccumulator — error — v4.0 |
| lsqvce:accum:unknownSystem | engine.BlockAccumulator — error — v4.0 |
| lsqvce:adjustBlock:badCell | engine.private.normalizeSystems — error — v4.0 |
| lsqvce:adjustBlock:badLoader | engine.private.normalizeSystems — error — v4.0 |
| lsqvce:adjustBlock:badStruct | engine.private.normalizeSystems — error — v4.0 |
| lsqvce:adjustBlock:badSystems | engine.private.normalizeSystems — error — v4.0 |
| lsqvce:adjustBlock:noDays | engine.private.normalizeSystems — error — v4.0 |
| lsqvce:adjustNonlinear:badX0 | engine.adjustNonlinear — error — v4.0 |
| lsqvce:adjustNonlinear:diverged | engine.adjustNonlinear — warning — v4.0 |
| lsqvce:adjustNonlinear:maxIter | engine.adjustNonlinear — warning — v4.0 |
| lsqvce:adjustNonlinear:nonFinite | engine.adjustNonlinear — error — v4.0 |
| lsqvce:adjustRobust:collapsed | engine.adjustRobust — error — v4.0 |
| lsqvce:adjustRobust:notConverged | engine.adjustRobust — warning — v4.0 |
| lsqvce:adjustVCEjoint:arcMismatch | engine.adjustVCEjoint — error — v4.0 |
| lsqvce:adjustVCEjoint:badAxisRows | engine.adjustVCEjoint — error — v4.0 |
| lsqvce:adjustVCEjoint:badC0 | engine.adjustVCEjoint — error — v4.0 |
| lsqvce:adjustVCEjoint:badSigma2_0 | engine.adjustVCEjoint — error — v4.0 |
| lsqvce:adjustVCEjoint:localNotPD | engine.adjustVCEjoint — error — v4.0 |

| identifier | where / what |
|---|---|
| `lsqvce:adjustVCEjoint:needCfcn` | engine.adjustVCEjoint — error — v4.0 |
| `lsqvce:adjustVCEjoint:noArcs` | engine.adjustVCEjoint — error — v4.0 |
| `lsqvce:adjustVCEjoint:notConverged` | engine.adjustVCEjoint — warning — v4.0 |
| `lsqvce:adjustVCEjoint:arcFactorsDiverging` | engine.adjustVCEjoint — warning — v5.3 |
| `lsqvce:adjustVCEjoint:notPD` | engine.adjustVCEjoint — error — v4.0 |
| `lsqvce:adjustVCEjoint:skipThreshold` | engine.adjustVCEjoint — error — v4.0 |
| `lsqvce:applyReg:badLambda` | engine.applyRegularisation — error — v4.0 |
| `lsqvce:applyReg:badRegMatrix` | engine.applyRegularisation — error — v4.0 |
| `lsqvce:applyReg:notPD` | engine.applyRegularisation — error — v4.0 |
| `lsqvce:applyReg:sizeMismatch` | engine.applyRegularisation — error — v4.0 |
| `lsqvce:applyReg:unknown` | engine.applyRegularisation — error — v4.0 |
| `lsqvce:ar2arma:degenerate` | kernel.ar2arma — error — v4.0 |
| `lsqvce:ar2arma:degenerateMA` | kernel.ar2arma — warning — v4.0 |
| `lsqvce:ar2arma:tooShort` | kernel.ar2arma — error — v4.0 |
| `lsqvce:ar2arma:unstable` | kernel.ar2arma — warning — v4.0 |
| `lsqvce:arcVarianceFactors:arcMismatch` | engine.arcVarianceFactors — error — v4.0 |
| `lsqvce:arcVarianceFactors:badMean` | engine.arcVarianceFactors — warning — v4.0 |
| `lsqvce:arcVarianceFactors:clamped` | engine.arcVarianceFactors — warning — v4.0 |
| `lsqvce:armaACV:nonStationary` | kernel.armaACV — error — v4.0 |
| `lsqvce:armaACV:zeroLeadingAR` | kernel.armaACV — error — v4.0 |
| `lsqvce:armaWhiten:acvTooShort` | kernel.armaWhiten — error — v4.0 |
| `lsqvce:armaWhiten:badSigma` | kernel.armaWhiten — error — v4.0 |
| `lsqvce:armaWhiten:sizeMismatch` | kernel.armaWhiten — error — v4.0 |
| `lsqvce:armaWhiten:startupNotPD` | kernel.armaWhiten — warning — v4.0 |
| `lsqvce:batchSystem:badMatrix` | engine.batchSystem — error — v4.0 |
| `lsqvce:batchSystem:badSource` | engine.batchSystem — error — v4.0 |
| `lsqvce:batchSystem:weightMatrix` | engine.batchSystem — error — v4.0 |
| `lsqvce:callUser:badHandle` | util.callUser — error — v4.0 |
| `lsqvce:callUser:threw` | util.callUser — error — v4.0 |
| `lsqvce:callUser:tooFew` | util.callUser — error — v4.0 |
| `lsqvce:checkPSD:badFit` | engine.checkPSD — warning — v4.0 |
| `lsqvce:checkPSD:badModel` | engine.checkPSD — error — v4.0 |
| `lsqvce:checkPSD:marginalFit` | engine.checkPSD — warning — v4.0 |
| `lsqvce:checkPSD:unstable` | engine.checkPSD — warning — v4.0 |
| `lsqvce:collectResiduals:badRes` | engine.collectResiduals — error — v4.0 |
| `lsqvce:covComponent:badLag` | engine.covComponent — error — v4.0 |
| `lsqvce:covComponent:lagTooLarge` | engine.covComponent — error — v4.0 |
| `lsqvce:covModel:badCep` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:badPsdMode` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:badSigma` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:badWhitenMode` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:cepOnVar` | engine.covModel — error — v5.3 |
| `lsqvce:covModel:degeneratePSD` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:negativePSD` | engine.covModel — warning — v4.0 |
| `lsqvce:covModel:noACF` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noAR` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noCovForm` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noFcn` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noFilterForm` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noPSD` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noSigma` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noType` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:noVAR` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:notWholeEpochs` | engine.covModel — error — v4.0 |

| identifier | where / what |
|---|---|
| `lsqvce:covModel:psdSanitationEscape` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:sigmaLength` | engine.covModel — error — v4.0 |
| `lsqvce:covModel:unknownType` | engine.covModel — error — v4.0 |
| `lsqvce:crossCorr:shortSeries` | kernel.crossCorr — warning — v4.0 |
| `lsqvce:datum:BnotNull` | engine.datum — warning — v4.0 |
| `lsqvce:datum:autoDetected` | engine.datum — warning — v4.0 |
| `lsqvce:datum:badB` | engine.datum — error — v4.0 |
| `lsqvce:datum:rankDeficientB` | engine.datum — error — v4.0 |
| `lsqvce:datum:zeroN` | engine.datum — error — v4.0 |
| `lsqvce:dctI:tooShort` | kernel.dctI — error — v4.0 |
| `lsqvce:decorrFactor:notPD` | kernel.decorrFactor — mixed — v4.0 |
| `lsqvce:diagnose:badPmatrix` | engine.diagnose — error — v4.0 |
| `lsqvce:diagnose:badTest` | engine.diagnose — error — v4.0 |
| `lsqvce:diagnose:batchMismatch` | engine.diagnose — error — v4.0 |
| `lsqvce:diagnose:largeU` | engine.diagnose — warning — v4.0 |
| `lsqvce:diagnose:nBatch` | engine.diagnose — error — v4.0 |
| `lsqvce:diagnose:noRobustScale` | engine.diagnose — error — v4.0 |
| `lsqvce:diagnoseAccum:needKeepLocal` | engine.diagnoseAccum — error — v4.0 |
| `lsqvce:diagnoseAccum:noSystems` | engine.diagnoseAccum — error — v4.0 |
| `lsqvce:ellipse:badPair` | engine.ellipse — error — v4.0 |
| `lsqvce:ellipse:samePair` | engine.ellipse — error — v4.0 |
| `lsqvce:fit:badArcs` | engine.declaration — error — v4.0 |
| `lsqvce:fit:badEstimate` | fit — error — v4.0 |
| `lsqvce:fit:badId` | engine.declaration — error — v4.0 |
| `lsqvce:fit:badMatrix` | fit — error — v4.0 |
| `lsqvce:fit:badParamColumns` | engine.declaration — error — v4.0 |
| `lsqvce:fit:badPrior` | fit — error — v4.0 |
| `lsqvce:fit:badScope` | engine.declaration — error — v4.0 |
| `lsqvce:fit:builderKind` | engine.declaration — error — v4.0 |
| `lsqvce:fit:duplicateSystem` | engine.declaration — error — v4.0 |
| `lsqvce:fit:lawOnFixedStructure` | fit — error — v4.0 |
| `lsqvce:fit:missingField` | engine.declaration;fit — error — v4.0 |
| `lsqvce:fit:modelKind` | fit — error — v4.0 |
| `lsqvce:fit:noCommon` | engine.declaration — error — v4.0 |
| `lsqvce:fit:noDays` | engine.declaration — error — v4.0 |
| `lsqvce:fit:noSystems` | engine.declaration — error — v4.0 |
| `lsqvce:fit:notYet` | fit — error — v4.0 |
| `lsqvce:fit:priorNotActive` | fit — warning — v4.0 |
| `lsqvce:fit:uCount` | fit — error — v4.0 |
| `lsqvce:fit:unknownField` | engine.declaration;fit — error — v4.0 |
| `lsqvce:ftest:differentData` | engine.ftest — error — v4.0 |
| `lsqvce:ftest:invalidPremise` | engine.ftest — warning — v4.0 |
| `lsqvce:ftest:noRedundancy` | engine.ftest — error — v4.0 |
| `lsqvce:ftest:notNested` | engine.ftest — error — v4.0 |
| `lsqvce:ftest:zeroVariance` | engine.ftest — error — v4.0 |
| `lsqvce:gcv:atLowerBound` | engine.gcvLambda — warning — v4.0 |
| `lsqvce:gcv:atUpperBound` | engine.gcvLambda — warning — v4.0 |
| `lsqvce:gcv:degenerate` | engine.gcvLambda — warning — v4.0 |
| `lsqvce:gcvLambda:qzFallback` | engine.gcvLambda — warning — v4.0 |
| `lsqvce:jointArcParts:sizeMismatch` | engine.private.jointArcParts — error — v4.0 |
| `lsqvce:jointModelOpt:badC` | engine.jointModelOpt — error — v4.0 |
| `lsqvce:jointModelOpt:badModel` | engine.jointModelOpt — error — v4.0 |
| `lsqvce:jointModelOpt:badStream` | engine.jointModelOpt — error — v4.0 |
| `lsqvce:leverage:outOfRange` | engine.leverage — warning — v4.0 |

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| lsqvce:leverage:sizeMismatch | engine.leverage — error — v4.0 |
| lsqvce:leverageBlocks:cNotPD | engine.leverageBlocks — error — v4.0 |
| lsqvce:leverageBlocks:lNotPD | engine.leverageBlocks — error — v4.0 |
| lsqvce:leverageBlocks:sizeMismatch | engine.leverageBlocks — error — v4.0 |
| lsqvce:levinson:notPD | kernel.levinsonDurbin — mixed — v4.0 |
| lsqvce:levinson:orderTooHigh | kernel.levinsonDurbin — error — v4.0 |
| lsqvce:localDiagnostics:notFinite | engine.localDiagnostics — error — v4.0 |
| lsqvce:log:noParallelQueue | util.log — warning — v4.0 |
| lsqvce:makeKeyFcn:badGroupBy | engine.private.makeKeyFcn — error — v4.0 |
| lsqvce:makeLoader:badArc | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badArcId | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badArcs | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:axisConflict | engine.makeLoader — error — v5.3 |
| lsqvce:makeLoader:badDay | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badDayCep | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badPrior | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badScope | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:badValidity | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:cepTwice | engine.makeLoader — error — v5.3 |
| lsqvce:makeLoader:buildDay | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:dayShape | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:localPlan | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:ncConflict | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:nlConflict | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:noCep | engine.makeLoader — warning — v5.3 |
| lsqvce:makeLoader:noArcs | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:noCov | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:noObsTypes | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:paramType | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:priorType | engine.makeLoader — error — v4.0 |
| lsqvce:makeLoader:rowWeightSize | engine.makeLoader — error — v4.0 |
| lsqvce:mapreduce:badTask | engine.mapreduce — error — v4.0 |
| lsqvce:mapreduce:serialFallback | engine.mapreduce — warning — v4.0 |
| lsqvce:normalEq:badMerge | engine.NormalEqSystem — error — v4.0 |
| lsqvce:normalEq:badWidth | engine.NormalEqSystem — error — v4.0 |
| lsqvce:normalEq:cancellation | engine.NormalEqSystem — warning — v4.0 |
| lsqvce:normalEq:noData | engine.NormalEqSystem — error — v4.0 |
| lsqvce:normalEq:noRedundancy | engine.NormalEqSystem — error — v4.0 |
| lsqvce:normalEq:notSPD | engine.NormalEqSystem — error — v4.0 |
| lsqvce:normalEq:sizeMismatch | engine.NormalEqSystem — error — v4.0 |
| lsqvce:prior:badTarget | engine.CommonPrior — error — v4.0 |
| lsqvce:probeMatrix:badDist | engine.private.probeMatrix — error — v4.0 |
| lsqvce:psd2acv:aboveNyquist | kernel.psd2acv — warning — v4.0 |
| lsqvce:psd2acv:nonPositiveVariance | kernel.psd2acv — error — v4.0 |
| lsqvce:psd2acv:tooFewPoints | kernel.psd2acv — error — v4.0 |
| lsqvce:psdVCE:arcMismatch | engine.psdVCE — error — v4.0 |
| lsqvce:psdVCE:degenerateStart | engine.psdVCE — warning — v4.0 |
| lsqvce:psdVCE:shapeMismatch | engine.psdVCE — error — v4.0 |
| lsqvce:qr:badMerge | engine.QRSystem — error — v4.0 |
| lsqvce:qr:badWidth | engine.QRSystem — error — v4.0 |
| lsqvce:qr:cancellation | engine.QRSystem — warning — v4.0 |
| lsqvce:qr:noData | engine.QRSystem — error — v4.0 |
| lsqvce:qr:noRedundancy | engine.QRSystem — error — v4.0 |
| lsqvce:qr:rankDeficient | engine.QRSystem — error — v4.0 |

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| `lsqvce:qr:singularR` | engine.QRSystem — error — v4.0 |
| `lsqvce:qr:sizeMismatch` | engine.QRSystem — error — v4.0 |
| `lsqvce:reduced:notSquare` | engine.ReducedContribution — error — v4.0 |
| `lsqvce:reduced:sizeMismatch` | engine.ReducedContribution — error — v4.0 |
| `lsqvce:redundancy:hutchFailed` | engine.redundancy — warning — v4.0 |
| `lsqvce:redundancy:hutchinsonExperimental` | engine.redundancy — warning — v4.0 |
| `lsqvce:redundancy:identityViolated` | engine.redundancy — warning — v4.0 |
| `lsqvce:redundancy:noStore` | engine.redundancy — error — v4.0 |
| `lsqvce:redundancy:smallGroup` | engine.redundancy — warning — v4.0 |
| `lsqvce:redundancy:spmdNoCollect` | engine.redundancy — warning — v4.0 |
| `lsqvce:redundancyLags:badEbar` | kernel.redundancyLags — error — v4.0 |
| `lsqvce:redundancyLags:badLocal` | kernel.redundancyLags — error — v4.0 |
| `lsqvce:redundancyLags:badUiZ` | kernel.redundancyLags — error — v4.0 |
| `lsqvce:redundancyLags:badW` | kernel.redundancyLags — error — v4.0 |
| `lsqvce:residuals:badSol` | engine.residuals — error — v4.0 |
| `lsqvce:residuals:noStore` | engine.residuals — error — v4.0 |
| `lsqvce:residuals:sigma0Mismatch` | engine.residuals — warning — v4.0 |
| `lsqvce:residuals:supersedesSol` | engine.residuals — warning — v4.0 |
| `lsqvce:resolveNc:conflict` | engine.resolveNc — error — v4.0 |
| `lsqvce:resolveNc:empty` | engine.resolveNc — error — v4.0 |
| `lsqvce:resolveNc:missing` | engine.resolveNc — error — v4.0 |
| `lsqvce:robustWeights:badInfluence` | kernel.robustWeights — error — v4.0 |
| `lsqvce:robustWeights:badTuning` | kernel.robustWeights — error — v4.0 |
| `lsqvce:runMeta:badArg` | engine.runMeta — error — v4.0 |
| `lsqvce:separable:acvTooShort` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:badSigma` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:badSigmaShape` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:epochNotPD` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:notWholeEpochs` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:sigmaLength` | kernel.separableWhiten — error — v4.0 |
| `lsqvce:separable:startupNotPD` | kernel.separableWhiten — warning — v4.0 |
| `lsqvce:sharedProbe:badProbes` | engine.sharedProbe — error — v4.0 |
| `lsqvce:sharedProbe:notSquare` | engine.sharedProbe — error — v4.0 |
| `lsqvce:snoop:badBatch` | engine.snoop — error — v4.0 |
| `lsqvce:snoop:badIndex` | engine.snoop — error — v4.0 |
| `lsqvce:snoop:breakdown` | engine.snoop — error — v4.0 |
| `lsqvce:snoop:contaminated` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:correlatedBatch` | engine.snoop — error — v4.0 |
| `lsqvce:snoop:downdateFailed` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:forcedBreakdown` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:forcedContaminated` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:forcedOverCap` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:lowRedundancy` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:maxReject` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:rejectFraction` | engine.snoop — warning — v4.0 |
| `lsqvce:snoop:tooContaminated` | engine.snoop — error — v4.0 |
| `lsqvce:stat:badDof` | engine.stat — error — v4.0 |
| `lsqvce:stat:badProb` | engine.stat — error — v4.0 |
| `lsqvce:stat:sizeMismatch` | engine.stat — error — v4.0 |
| `lsqvce:streamContribution:badSol` | engine.streamContribution — error — v4.0 |
| `lsqvce:streamContribution:cNotPD` | engine.streamContribution — error — v4.0 |
| `lsqvce:streamContribution:noArcs` | engine.streamContribution — error — v4.0 |
| `lsqvce:streamContribution:noStore` | engine.streamContribution — error — v4.0 |
| `lsqvce:synthLoader:badArc` | engine.synthLoader — error — v4.0 |

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| `lsqvce:synthLoader:noNoiseModel` | engine.synthLoader — error — v4.0 |
| `lsqvce:systematic:badCovariate` | engine.systematic — error — v4.0 |
| `lsqvce:systematic:lbUnreliable` | engine.systematic — warning — v4.0 |
| `lsqvce:systematic:noResiduals` | engine.systematic — error — v4.0 |
| `lsqvce:systematic:tooFew` | engine.systematic — warning — v4.0 |
| `lsqvce:toeplitzCofactor:badVariance` | kernel.toeplitzCofactor — warning — v4.0 |
| `lsqvce:toeplitzCofactor:empty` | kernel.toeplitzCofactor — error — v4.0 |
| `lsqvce:toeplitzCofactor:notACV` | kernel.toeplitzCofactor — warning — v4.0 |
| `lsqvce:traceEstimate:badOperator` | engine.traceEstimate — error — v4.0 |
| `lsqvce:traceEstimate:needSize` | engine.traceEstimate — error — v4.0 |
| `lsqvce:traceEstimate:notSquare` | engine.traceEstimate — error — v4.0 |
| `lsqvce:traceEstimate:sizeMismatch` | engine.traceEstimate — error — v4.0 |
| `lsqvce:util:log:badKey` | util.log — error — v4.0 |
| `lsqvce:varACV:badA` | kernel.varACV — error — v4.0 |
| `lsqvce:varACV:notStationary` | kernel.varACV — error — v4.0 |
| `lsqvce:varWhiten:badA` | kernel.varWhiten — error — v4.0 |
| `lsqvce:varWhiten:badSigma` | kernel.varWhiten — error — v4.0 |
| `lsqvce:varWhiten:badStartup` | kernel.varWhiten — error — v4.0 |
| `lsqvce:varWhiten:notWholeEpochs` | kernel.varWhiten — error — v4.0 |
| `lsqvce:vce:badAccFcn` | engine.vce — error — v4.0 |
| `lsqvce:vce:badSigma0` | engine.vce — error — v4.0 |
| `lsqvce:vce:groupMismatch` | engine.vce — error — v4.0 |
| `lsqvce:vce:hutchGroupMismatch` | engine.vce — error — v4.0 |
| `lsqvce:vce:lowRedundancy` | engine.vce — warning — v4.0 |
| `lsqvce:vce:notConverged` | engine.vce — warning — v4.0 |
| `lsqvce:verbosity:badLevel` | util.verbosity — error — v4.0 |
| `lsqvce:verbosity:badTag` | util.verbosity — error — v4.0 |
| `lsqvce:whittle:badG` | kernel.whittle — error — v4.0 |
| `lsqvce:whittle:notPD` | kernel.whittle — error — v4.0 |